# Prospective validation of AI-based scoring functions in drug discovery

by

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# Abbreviations

| | |
|---|---|
| AI | Artificial intelligence |
| ML | Machine learning |
| GCI | Grafting Coupled Interferometry |
| HTS | high-throughput screening |
| vHTS | virtual HTS |
| CADD | Computer Aided Drug Design |
| SBDD | Structure Based Drug Design |
| LBDD | Ligand Based Drug Design |
| NMR | Nuclear Magnetic Resonance |
| MD | Molecular dynamics |
| SVM | support vector machines |
| CNN | convolutional neural networks |
| GNN | graph neural networks |
| PDB | Protein Data Bank |
| DUD | Directory of Useful Decoys |
| Pa | Pseudomonas aerginosa |
| FabF | 3-oxoacyl-[acyl-protein] synthase 2 |
| PTM | Platensimycin |
| Hit 1 | 8R1V ligand |
| Hit 2 | 8R0I ligand |
| Hit 3 | 8PJ0 ligand |
| 1$^{st}$ gen EOS series | Hit 1, Hit 2, Hit 2 |
| EFSL | European Fragment Screening Library |

# Summary:

The development of new pharmaceutical drugs to treat diseases is both costly and time-consuming. A contributor to the cost and development time is associated with the discovery of novel drug candidates. Even though the number of molecules with drug-like features is estimated to be $10^{32}$ molecules, *in vitro* binding assays to asses binding kinetics can only screen a minute fraction of the total number of possible binders. The addition of in silico virtual screening of digital chemical libraries offers a faster but less accurate approach to finding novel binders. In short, virtual screening employs computational scoring methods to estimate the strength of binding between a small molecule and its proposed target. The rapid prediction of binding characteristics for millions of small molecules against a target often comes at the expense of prediction accuracy.

Recent advances in artificial intelligence (AI) have raised optimism regarding its potential applications in biomedical research. Machine learning (ML) which is a branch of AI, utilises statistical and mathematical methods to identify patterns and relationships within data for predictive purposes. In drug discovery, there is a growing interest in whether ML-based scoring functions can improve the prediction of binding affinity compared to currently employed scoring functions. By learning complex relationships from large datasets, ML-based methods may capture interaction patterns that are not adequately represented in classical scoring functions, potentially improving predictive accuracy while maintaining computational efficiency.

In recent years, multiple ML-based scoring functions have been developed for use in virtual screening and structure-based drug discovery. Computational benchmarking studies have demonstrated promising predictive performance for several of these models. However, despite encouraging benchmark results, ML-based scoring functions remain insufficiently evaluated in prospective drug discovery settings. In this thesis, six ML-based scoring functions were deployed in a virtual screening campaign to identify novel binders for the C164A mutant of Pseudomonas aeruginosa FabF (*Pa*FabF C164A) from a manually assembled virtual library of purchasable compounds. The six ML-based scoring functions were AK-Score2, KDeep, OnionNet2, Boltz-2, iScore and vScreenMLv2.0.

Each ML-based scoring function ranked small molecules of an assembled dataset based on their predicted binding strength towards the protein target. Of the top1% of ranked compounds, five compounds from each scoring function were selected for further in vitro affinity measurements by Grafting Coupled Interferometry (GCI). Additionally, five compounds were selected by visual inspection of poses generated by

molecular docking. Due to overlap between the selection of five molecules per scoring function and visual inspection, a total of 30 compounds were selected.

Of the selected compounds, 28 were evaluated by single point GCI measurements, identifying 20 binders. Subsequently, eight compounds were subjected to co-crystallization with PaFabF C164A, and X-ray crystallography. The crystal structure of two ligand-protein complexes were successfully resolved. The two small molecules SN-1221 and SN-0709 observed in the crystal had a measured dissociation constant Kd rate of 8.7μM and 39.2μM respectively.

The five ML-based scoring functions which either required a pose for binding prediction, or generated their own binding pose, all predicted poses within 2 Å of RMSD compared to the crystal pose of the two resolved ligand-protein complexes. Despite generating a pose within 2Å RMSD compared to the crystal pose, a key interaction was lost for both the predicted SN-1221 and SN-0709 pose generated by the Boltz-2 scoring function.

Despite a generally high level of disagreement in how the six ML-based scoring functions ranked the virtual library, all models were able to identify binders in their top 1% rank. AK-Score2 and KDeep exhibited the strongest performance as they identified potent binders while also achieving a 100% hit rate among their selected compounds.

These results demonstrate that ML-based scoring functions can support the identification of novel binders. However, their predictive performance remains highly model dependant highlighting the importance of complementing predictions with experimental validation.

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# 1.Introduction

## 1.1. Drug Discovery

Drug discovery and development is a multidisciplinary field aimed at discovering and developing novel pharmaceutical compounds to treat, prevent or manage disease(Klebe, 2024). The drug discovery and development pipeline consists of several phases, each presenting unique objectives and challenges. These phases include target validation, assay development, hit and lead identification, candidate selection, preclinical testing, clinical trials, and finally regulatory approval. The full development of a new drug typically requires 12–15 years and costs that on average exceed 2 billion dollars. Although numerous advances in drug discovery strategies have been made over recent decades, the majority of drug candidates still fail before reaching the market. Approximately nine out of ten candidates do not successfully pass clinical trials or obtain regulatory approval, and the overall failure rate is even greater when preclinical stages are considered (Sun et al., 2022). Given the extensive time and financial investments required for drug development, these challenges emphasize the importance of improving how potential drug candidates are identified and prioritized early in the discovery process.

### 1.1.1 Hit discovery

Once a target has been identified, the next stage in drug discovery include the identification of novel molecules that exhibit desired activity against the target (Janzen, 2014; Klebe, 2013). Active molecules identified through biological assays are referred to as hits. Hits are primarily used to identify which interactions in the binding site contribute to a stronger binding. From good hit candidates, a hit expansion can be performed to find new interaction that eventually will lead to compound with lead characteristic (Sliwoski et al., 2014). Compounds exerting close to favourable affinities are referred to as lead candidates (Klebe, 2013)

A common approach for discovering novel hits against a target is the screening of chemical libraries using high-throughput screening (HTS). HTS is an assay-based approach that enables the rapid testing of large compound libraries for activity against a target (Hansen, 2025). Despite enabling the search for hits on a relatively large scale, HTS is constrained by factors such as cost, infrastructure requirements, and assay capacity, which limit the number of compounds that can realistically be tested experimentally (Roy, 2021). Advances in computer technology over the past decades and their implementation in cheminformatics have enabled the expansion of molecular collections into increasingly large virtual libraries (van Hilten et al., 2019). Increasing the size of a chemical library used for screening can potentially yield a greater number of hits and thus improve the likelihood of identifying potent binders. As a result, the process of discovering and developing new drug candidates could become more efficient (Gloriam, 2019). With the emergence of ultra large virtual chemical libraries consisting of millions to billions of drug like compounds (Carlsson & Luttens, 2024), HTS acts as a bottleneck as it realistically will not be able to screen huge libraries (Roy, 2021).

## 1.1.2 Computer Aided Drug Design (CADD)

The implementation of computational approaches in the early stages of drug discovery such as hit identification and expansion, gained popularity in both industry and academia for their potential to reduce reliance on physical experiments during the early 21st century (Shoichet, 2004). CADD includes two major approaches which differ in their usage based on the availability of target information. These approaches include Structure Based Drug Design (SBDD) and Ligand Based Drug Design (LBDD). When information of target structure is known through methods such as X-ray crystallography and Nuclear Magnetic Resonance (NMR) spectrocopy, a SBDD approach can be utilised to predict new binders towards a target. The core hypothesis of this approach is that molecules which favourably interact with the target, will modulate the targets biological function. Discovering novel binders which could favourable interact with the target, can be elucidated by careful analysis on the structural information of the target binding site (Sliwoski et al., 2014).

Proteins are the most common class of drug targets because they perform essential biological functions which includes catalysis, signal transduction, transport, and a wide range of other cellular processes (Bull & Doig, 2015; Kim, 2023). Protein-ligand interaction are governed by the change in free energy. The change in free energy of binding is composed by both enthalpic and entropic contributions which are related to the equilibrium constant through the expression:

$$\Delta G = \Delta H - T\Delta S = -RTlnK$$

In protein-ligand binding, ΔG represents the change in binding free energy whereas the change in enthalpy (ΔH) is associated with the formation and breakage of bonds during binding. TΔS represents the change in entropy in relation to the temperature and is associated with the overall disorder of the system which includes alterations in molecular flexibility, translational and rotational freedom, and solvent displacement (Bronowska, 2011). The major contributors to entropy during binding are the displacement of water molecules from the protein binding pocket and the loss of conformational freedom of both the ligand and the protein upon complex formation (Sliwoski et al., 2014).

SBDD methods such as molecular docking, virtual HTS (vHTS) and Molecular dynamics (MD) simulations try to some extent to predict the change in free energy upon binding a small molecule to a known protein. MD simulations use extensive data processing to predict protein-ligand interactions in a dynamic setting where the system is allowed to move. Due to the heavy computational power required to predict the dynamic relationship, it is better suited for the continued optimisation of already promising lead candidates (Salo-Ahen et al., 2020; Sliwoski et al., 2014).

## 1.1.3 vHTS and molecular docking

In vHTS, the goal is to find hit candidates from large datasets. Thus it is necessary to strike a balance between predicting the most important factors for binding while not requiring too much computational power which would increase the prediction duration. An approach to this is to predict already generated placements of the molecules from the chemical library. The ability to Molecular docking is the approach to generate plausible placements of the potential ligands in the binding pocket. These placements are referred to as poses. These poses are generated by search algorithms which systematically try to find possible ligand confirmations, orientations and positions within the binding pocket while allowing some ligand and protein flexibility (Xuan-Yu Meng, 2011). In vHTS, molecular docking engines are used to generate poses for all small-molecules used in the virtual chemical library. From the generated poses, scoring functions try to predict the binding affinity of a ligand to a target binding site based on the physicochemical properties of the ligand and the protein (Sliwoski et al., 2014). Scoring functions are often integrated into molecular docking engines where they are used to both evaluate and rank the generated ligand poses based on predicted binding affinity. The ability of a scoring function to identify poses close to the experimentally observed binding pose, is referred to as docking power (Agu et al., 2023).

### 1.1.4 Classical scoring functions

Historically, there have been three different approaches to how scoring functions predict protein ligand interactions. (1) Force field based scoring functions model physical interactions such as electrostatic interactions, Van der Waals and bonded terms to calculate the change in free energy for the complex formation. The physical parameters are derived from experimental data. Force field scoring functions can biased by favouring charged ligands. (2) Empirical scoring functions estimates the binding affinity based on a set of weighed parameters. The weight is based on how much each parameter has contributed to binding for a set of protein-ligand complexes in the training data. The parameters represent physical properties such as bond length, hydrophobic contacts, entropy changes and desolvation effects. Due to the more simple energy terms, they are much faster than force-field based scoring functions. (3) Knowledge based scoring functions predict the affinity between a ligand and a protein binding pocket by comparing the statistical interpretations of which types of interactions are more frequent in protein-ligand structures in the training data, with the test-complex (Huang et al., 2010). These types of scoring functions haves seen wide use in CADD and are highly integrated in the general drug discovery workflows (Sliwoski et al., 2014). However, due to the simplified physics and empirical parametrization, classical scoring functions are subjected to certain biases. They often favour larger, more hydrophobic and more flexible ligands while inaccurately predicting entropic factors such as de-solvation (Gregory L. Warren et al., 2006).

## 1.2 Artificial Intelligence (AI) in hit discovery

With the emergence of artificial intelligence-based approaches, a new type of scoring functions has emerged. The traditional CADD approach of using classical scoring functions for virtual screening, struggles to strike a balance between accurate native prediction and short processing time. Traditional scoring functions are incomplete in the sense that they oversimplify the complex binding process. Drug discovery could benefit from a less linear and distinctively different approach to interpreting and more accurately predict the binding process. The increase in popularity of Machine learning (ML) in recent years has given rise to a novel ML-based scoring function class as an extension of empirical scoring functions (Harren et al., 2024). ML, a subfield of AI involves the development of algorithms which learn patterns and relationships from data to improve task performance through experience (Courville, 2016). This differs from classical scoring functions where factors correlating with strong binding are predetermined in static algorithms. ML based scoring functions try to extract a more detailed relationship by learning from the underlying data. As ML based scoring functions consistently have demonstrated to outperform all types of classical scoring functions on standard benchmarks (Harren et al., 2024), there is optimism for real world applications

### 1.2.1 The architecture of ML-based scoring functions

There are several different ways to structure how a ML-based scoring function interprets and predicts protein-ligand binding. The different architectures employed among ML based scoring functions are often highly complex and hard to interpret. In general terms, ML is categorised into classical approaches and neural networks. Classical machine learning methods use engineered descriptors which describes the protein-ligand complex, as input in learning algorithms such as such as linear regression, decision trees, and support vector machines (SVM) (Harren et al., 2024).

Neural network based ML has seen massive progress in fields such as image processing and natural language processing (Harren et al., 2024). These models are composed of a network of single components called neurons. Each neuron is a simple linear regression model paired with a non-linear activation function. The latter allows a neural network to learn irregular relationships that allow the model to capture more complex pattern that does not follow linearity. By organizing neurons into multiple layers, deep neural networks can extract relevant features and learn complex nonlinear relationships directly from data (Harren et al., 2024)

Within neural network based scoring functions, the two classes convolutional neural networks (CNN) and graph neural networks (GNN) are especially relevant(Harren et al., 2024). CNN based scoring functions interpret protein-ligand interactions by filtering information from multi-dimensional grids enclosing the binding pocket (Cun L, 1989; Harren et al., 2024). In contrast, GNN based scoring functions operate on graph structure data where atoms and bonds within the binding pocket are represented as nodes and edges. GNN models learn by updating each graph section based on its relation to neighbouring nodes and edges (Zhou et al., 2020).

### 1.2.2 Training datasets

As machine learning models make predictions based on previously seen data, they are optimised to perform predictions on similar data. This works well in fields where the test data closely resembles training data such as image recognition. However, in chemical applications such as protein-ligand binding, the vast and diverse chemical space means that molecules encountered during application often differ significantly from those in the training data . Most ML-based scoring functions are trained on datasets made to test and compare classical scoring functions (Harren et al., 2024). These benchmark datasets were never intended to be used for training models. This pose the question of how effective ML based scoring functions are at actually identifying possible hits. In an analysis performed by (Harren et al., 2024) on the effectiveness of learning from similar and dissimilar data, there was a trend of ML scoring functions only outperforming empirical scoring functions when they met highly similar structures to training data structures. There is much evidence that available benchmark datasets are not capable of assessing the performance of ML scoring functions in close to real world scenarios due to the bias of high similarity to test-data. In principle ML based scoring functions should be capable to learn the physics of protein-ligand binding, but this depends to on the size and quality of the training dataset (Harren et al., 2024).

The dataset PDBbind (Renxiao Wang, 2005)has seen a wide use in training of ML-based scoring functions (Harren et al., 2024). PDBbind contains structural data from the Protein Data Bank (PDB) (Vallat et al., 2026) linked to binding affinity data from the original literature. The subset refined PDBbind conists of high quality protein structures (Renxiao Wang, 2005).

While datasets such as PDBbind contain experimentally determined binding modes with affinities, they often lack explicit examples of inactive molecules that do not bind. Without any examples of inactive molecules, the models might be trained to assume that every molecule will bind thus making it difficult to distinguish active molecules from inactive molecules when encountering novel data. A key challenge with training ML-based models on negative data is that it is the difficulty of obtaining an experimentally determined structure of an inactive or weak binder. An approach to solve the lack of true negative data is by computationally generating decoy molecules assumed to be at the best weak binders. Such decoy structures can be found in the Directory of Useful Decoys (DUD). In the DUD datasets DUD, DUD-E and DUD-Z, decoy molecules are generated by selecting molecules with the same physiochemical features as a strong binder but with a different chemical structure (Mysinger et al., 2012).

### 1.2.3 Specialisation of scoring functions and reliance on training data

The key to understanding how a ML based scoring function works lies in the training data. Traditionally, ML methods work under a simple assumption that training and test or application data are drawn from the same underlying distribution. The performance of ML methods is inherently limited if the representation does not contain sufficient information to describe the observed relationships. Conversely, highly complex representations might contain many features not sufficiently represented in the provided data, making the model much more susceptible to learning biases. The sensitivity of a model towards solving problems close to the data it has been trained on, is a key factor for why ML-based models tend to be more specialised (Harren et al., 2024).

Traditionally, one scoring function has been used to perform a variety of tasks such as selecting most probable pose, identifying active compounds in virtual screening and ranking potent binders in the lead optimisation stage. This differs from ML-based functions which typically tend to be oriented towards solving one task based on what it has seen in the training data (Harren et al., 2024).

### 1.2.4 Overview of the scoring functions used in this thesis

With the large variety of published novel ML based scoring functions over the recent years, it is difficult to assess the advantages of each individual functions (Harren et al., 2024). In this thesis, the performance of six different ML-based scoring functions was investigated. It was chosen to select ML-scoring functions with differences in architecture and training as it would allow for a possible comparison between which ML-scoring function design approach could work the best for hit expansion for PaFabF C164A. According to (Harren et al., 2024), it should be possible to use ML-based scoring functions with different task specialisations such as predicting the correct pose, predicting an accurate binding affinity prediction and filtering true binders from non-binders, for virtual screening. The six ML based scoring functions was also selected based on their stated benchmarks and ease of use.

KDeep: The scoring function KDeep is based on a 3D CNN architecture. The model was trained on a refined subset of protein-ligand complexes from PDBbind (v.2016) consisting of complexes with a resolution lower than 2.5Å, R-factor lower than 0.25, non-covalent ligand binding, no steric clashes or ternary complexes and with Kd or Ki values > 1pM (Jimenez et al., 2018).

vScreenML v2.0: The model is based on the extreme gradient boosting framework, a decision tree algorithm aimed at classification. vScreenML v2.0 provides a value between 1 and 0 based on the likelihood of the compound being an active binder. The closer the value is to 1, the more likely the compound is a binder. The model was trained on 1806 active structures and 4475 decoy structures generated by DUD-E (Andrianov et al., 2024).

Ak-Score2: The model predicts the change in free energy of a protein-ligand interaction by combining the inputs of three sub-networks (AK-Score-NonDock, AK-Score-DockS and AK-Score-DockC) and a physics based scoring functions. The three sub-networks use a GNN architecture, with each serving a distinct function. AK-Score-NonDock considers features of the ligand and protein binding pocket, rather than the pose to classify whether the ligand is active or not. AK-Score-DockS and AK-Score-DockC both take poses into consideration when predicting binding strength. They both calculate RMSE of the predicted score versus experimental binding affinities. The DockS differ from DockC by predicting RMSE variation and predicted free energy of binding as separate values. The models were trained on four datasets: Native set, conformational decoys set, cross docked decoys set and random decoys set. The native set contained structures from PDBbind v.2020. The conformational decoy set consisted of redocked poses by AutoDock of the native set. The cross docked set contained AutoDock generated complexes of ligands from other structures docked with new structures in the native set. The last decoy dataset consisted of ligands of the Enamine Discovery set (DDS-50) docked with protein structures of the native set (Hong et al., 2024).

iScore: The iScore scoring function is based on an ensemble of three models with different ML architecture: Deep neural networks (iScore-DNN), random forest (iScore-RF), extreme gradient boosting (iScore-XGB). iScore Hybrid predicts based on the three individual base learners. iScore does not require a pose as it predicts the binding affinity (pKd) based on a set of descriptors calculated from the SMILES of the ligand and the PDB file of the protein. iScore was trained on the PDBbind 2020 refined set (Mahdizadeh & Eriksson, 2025).

OnionNet2: OnionNet2-2 is a scoring function based on a 2D CNN. It compresses the 3D geometry of a protein-ligand complex into a 2D distance-based contact map. The model was trained on 11,909 complexes derived from the PDBbind v2016 dataset, excluding complexes present in PDBbind refined and core set. The predicted binding strength is outputted as binding affinity in pKa scale (Wang et al., 2021)

Boltz-2: This is a structure prediction model that utilises a deep neural network to generate protein-ligand complexes to predict native poses and binding affinities. The Boltz-2 structure model was trained on experimental data from PDB v.2023-06-01, molecular dynamics snapshots from MISATO(Till Siebenmorgen et al., 2024), ATLAS(Yann Vander Meersche, 2024) and (Antonio Mirarchi, 2024) CATH open efforts and high confidence structures from Alphafold2 and Boltz1. For affinity prediction, sound regression values (e.g Ki, Kd, IC50, EC50, XC50) for single protein targets were gathered from Pubchem, ChEMBL and BindingDB. For binary affinity classification, both experimental affinity data from Pubchem HTS (>100 compound, hit rate <10%), the CeMM fragment screening dataset and the MIDAS protein-metabolite dataset, was combined with synthetic decoys created from shuffled binders with a Tanimoto similarity <0.3. For screening, Boltz 2 requires the protein sequence and ligand as SMILES or SDF as input. It outputs a co-folded structure of the protein-ligand complex as a PDB file and a predicted general value of binding strength which can be approximately interpreted as an IC50-like value (Passaro et al., 2025).

## 1.3 *Pa*FabF as a target

The target selected for validating the scorings in this study was Pseudomonas aerginosa (*Pa*) 3-oxoacyl-[acyl-protein] synthase 2 C164A mutant. FabF catalyse the Claisen condensation of malonyl-ACP(acetyl carrier protein) with acyl-ACP in the bacterial fatty acid synthesis FASII pathway (J. Yao, 2017; L. Bibens, 2023). The first step in the reaction is the formation of a covalent bon between the negatively charged Cysteine 164 and an activated fatty acid to form a thioester. Next the thioester reacts with malonyl-ACP which gives β-ketoacyl-ACP whereas two more carbon atoms have been added to the original fatty acid chain (J.G. Olsen, 2001).

The FASII pathway is distinct from the FASI pathway in mammals, but highly conserved among bacterial pathogens (J. Yao, 2017). Inhibition of the FASII, depletion of lipid synthesis has shown to restore pathogenic sensitivity to colistin in previous colistin resistant systems (L.A. Carfrae, 2023). A potential inhibitor of FabF could thus be an antibiotic candidate. However as FASII is also present in the mitochondria of human cells, there is a possibility of a potential FASII drug affecting the host. On the other hand, preclinical screening trials has indicated a short-term toleration of a mitochondrial FASII inhibition (S.M. Nowinski, 2020).

## 1.3.1 Known FabF inhibitors

The current most potent inhibitor of FabF Platensimycin (PTM), has an activity in the nanomolar range towards the C164A version mimicking the reaction intermediate state of FabF. In vitro binding of PTM to the wildtype FabF displaying the apo state of a free cysteine, is much weaker (J. Wang, 2006; J.G. Olsen, 2001). The inability of PTM to access the target due to poor membrane permeability opens the search for other compounds with similar physicochemical and structural features scaffolds that may bind the same way(Charis Georgiou, 2025). A screening of the fragment library European Fragment Sceening Library (EFSL) against the PaFabF C164A variant found a hit in the low micromolar Kd range (35uM) referred to as Hit 1. A follow up screening of molecules containing the same core dimethyl-phenyl-pyrazolone with secondary amide scaffold as the initial hit found two additional hits in the low micromolar Kd range (Hit 1 and Hit 2) (Jalencas et al., 2024). Resolved structures of the binding mode of the three hits further validated the conserved dimethyl-phenyl-pyrazolone as important for binding. Key interactions of the conserved scaffold shown in Figure1.1 includes hydrogen bond between ligand secondary amide and the two histidine's His341 and His 304, backbone hydrogen interaction between ligand pyrazolone oxygen and Ala164 and Phe400, π–π interactions between ligand phenyl-pyrazolone and Phe400&Phe203 (Jalencas et al., 2024).

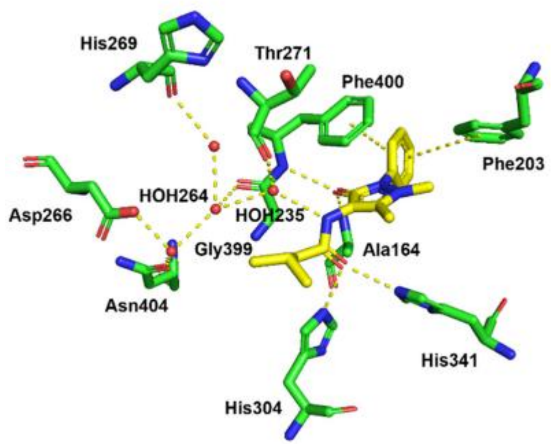



Figure1.1: The binding mode of Hit 1 in the binding pocket of *Pa*FabF C164A. Reproduced from (Jalencas et al., 2024).

## 1.3.2 Hit expansion of PaFabF:

The resolved structures of the three hits found the screening campaign against PaFabF wth EFSL fragment library, has contributed to a detailed understanding of key interactions in the malonyl binding pocket. It should be noted that Hit 1, Hit 2 and Hit 3 does not hold biological relevance as the potent binders of the C164A FabF version does not hold biological relevance as the dimethyl-phenyl-pyrazolone with the secondary amide does not bind to the FabF wild type. However, as the Hit 1, Hit 2 and

Hit 3 has commercially available analogues it serves as a good target for validating the six ML-based scoring functions. Furthermore, discovery of novel hits could lead to further information of key interactions in the PaFabF C164A malonyl binding pocket. In this project, Hit 1, Hit 2 and Hit 3 are referred to as the 1st gen EOS series.

## 1.4 Grating-Coupled Interferometry (GCI

GCI was used to measure binding of ligands to PaFabF C264A. GCI is an optical technique which offers a label free real time detection of molecular interactions (Jankovics et al., 2020; Malvern-Panalytical, 2026).

GCI is based on detecting small variations in the refractive index of a sensor by interferometry (Kozma et al., 2009). The refractive index of a medium such as a sensor coated with a specific protein, is determined by the local dielectric environment created by its molecular composition (Hecht, 2017). Changes in the refractive index can be measured through their effect on the phase of light travelling close to the sensor surface (Kozma et al., 2009). In GCI, a measurement beam is guided through a waveguide integrated along the sensor surface (Kozma et al., 2009; Malvern-Panalytical) After propagating along the sensor, the measurement beam is combined with a reference beam to generate an interference signal enabling detection of the induced phase shift. Ligands binding to proteins attached to the sensor surface alters the local dielectric environment causing a refractive index change (Jankovics et al., 2020) Variations in the refraction leads to an altered phase shift compared to the unbound state. Real time detection of phase-shift changes enables binding to be quantitatively described by association and dissociation kinetics (Hecht, 2017).

## 1.5 X-ray crystallography

X-ray crystallography is a biophysical method used to resolve the three dimensional atomic structure of a crystalline solid (*X-ray Crystallography*). In drug discovery, X-ray crystallography is commonly used to determine the binding mode of a ligand to its target molecule. During hit expansion and discovery, resolving the binding mode provides insight into key molecular interactions, which can guide the discovery of additional binders and/or further optimization of promising candidates (Blundell & Patel, 2004). A requirement for resolving a protein-ligand complex by X-ray crystallography, is the formation of a suitable crystal of the complex. A protein-ligand crystal is an ordered assembly of identical molecular complexes arranged in a periodic three-dimensional lattice (Drenth, 2007). Once a suitable crystal has been obtained, it is exposed to an X-ray with a wavelength of roughly 1Å. When the X-ray electromagnetic wave encounters electrons, the wave scatters. The highly ordered and periodic arrangement of atoms inside the crystal leads to the waves interfering constructively at specific angles, defined by Braggs Law, which produce discrete diffraction spots on a detector. The detected diffraction pattern represents the arrangement of atoms within the crystals. From the diffraction data, each spot can be indexed, integrated, merged and scaled by a computer to produce a reciprocal space-representation of the crystal. In order to fit the computed data into an interpretable electron density map, the phase problem must be solved. An X-ray wave is characterized by its amplitude, which determines the intensity of the diffraction, and its phase, which describes the relative position of the wave after scattering. The phase information is not captured during the physical measurement due to the quantum mechanical nature of X-rays and electrons. If a homologous protein exists, it can help solve the phase problem by providing a reference electron density map in a process called molecular replacement ("X-ray Protein Crystallography,")

# Aim

The aim of this thesis was to evaluate the performance of six machine learning (ML) based scoring functions to in a real world drug discovery scenario. As drug target, PaFabF C164A was selected as it was experimentally accessible and had a validated binding scaffold for which commercially available analogues existed.

To identify novel binders for PaFabF C164A through structure-based virtual screening, a curated compound library was screened *in silico*, and top-ranked compounds predicted by the different ML-based scoring functions were selected for prospective experimental validation. Binding affinities were measured using GCI, and binding mode were determined by X-ray crystallography. The generated experimental data served as basis upon which the performance of the different scoring functions was compared.

# Materials

## 2.1 Assembly of ENAMINE-EOS set

| SMARTS | Chemical name | Referred to as |
|---|---|---|
| *CN1N(C(=O)C(=C1C)NS(=O)(=O)[*:2])[a;r5,r6]* | N-sulfonyl-4-aminoantipyrine | Dimethyl-phenyl-pyrazolone with sulfonamide. |
| *CN1N(C(=O)C(=C1C)NC(=O)[*:2])[a;r5,r6]* | N-acyl-4-aminoantipyrine | Dimethyl-phenyl-pyrazolone with amide scaffold |

## 2.2 Compound preparation

**Brenk Lab scripts**

- Charges.py (Brenk, 2018a).
- Cl_tautomerizer.py (Brenk, 2018b).

Maestro LigPrep (Schrödinger Release 2021-2).

## 2.3 Molecular Docking

| PDB structures: | Assembly |
|---|---|
| 8R0I(V. Yadrykhinsky, Brenk, R, 2023) | Biological Assembly 1 |
| 8PJ0(Georgiou, 2023b) | Biological Assembly 1 |
| 8R1V(V. Yadrykhinsky, Brenk, R., 2023) | Biological Assembly 1 |

**Software package**: Open Babel 3.1.1

**Docking software**

Maestro (Schrödinger Release 2024-1), Glide docking engine. (source

SesSar (14.2) FexX docking engine. (Friesner et al., 2004)

## 2.4: Virtual Screening

Computer specs for virtual screening by AK-Score2 and vScreenMLv2.0:

| Operating system: | Processor: | Memory: |
|---|---|---|
| Ubuntu 24.04.4 LTS | Interl Core i7-4790*8 | 16GB |

**Kdeep**
Accessed: 14.12.25-18.12.25
Web GUI: Playmolecule.ai

**vSscreenMLv2.0**
Accessed: 16.12.25-17.12.25
Dependencies (conda venv): setup.py performed automatic installation of all reuired packages.

**AK-Score2**
Accessed: 16.12.25-19.12.25
Dependencies (python venv):
meeko 0.3.3. numpy 1.23.5, openbabel-wheel 3.1.1.22, pandas 1.5.3, rdkit-pypi 2022.3.1, torch 1.11.0+cpu, torch-cluster 1.6.0, torch_geometric 2.0.4, torch-scatter 2.0.9, torch-sparse 0.6.13, torch-spline-conv 1.2.1. torchaudio 0.11.0+cpu, torchvision 0.12.0+cpu.

Computer specs for virtual screening by iScore, OnionNet2 and Boltz-2

| Operating system: | Processor: | Memory: | GPU: |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Ubuntu 24.04.4 LTS | Intel Core i9-14900K | 62GB | NVIDIA RTX 4090 GPU |

**iScore**
Accessed: 17.12.25-20.12.25
Dependencies (conda venv):
python==3.11.4, ipykernel, numpy==1.26.4, conda-forge::scipy==1.11.2, nvidia/label/cuda-12.0.0::cuda, fpocket, pytest, rdkit

**OnionNet22**
Accessed: 17.12.25-20.12.25
Dependences (conda venv):
conda-forge::tensorflow-gpu, pandas, scikit-learn, openbabel, ipykernel, conda-forge::rdkit.

**Boltz-2**
Accessed: 18.12.25-21.12.25
Dependencies (python venv): python=3.10, ipykernel, matplotlib, pip, boltz==2.2.0
Protein sequence:
GHMASMSRRRVVITGMGMLSPLGLDVPSSWEGILAGRSGIAPIEHMDLSAYSTRFGGSVKGFN
VEEYLSAKEARKLDLFIQYGLAASFQAVRDSGLEVTDANRERIGVSMGSGIGGLTNIENNCRSLF
EQGPRRISPFFVPGSIINMVSGFLSIHLGLQGPNYALTTAATTGTHSIGMAARNIAYGEADVMVA
GGSEMAACGLGLGGFGAARALSTRNDEPTRASRPWDRDRDGFVLSDGSGALVLEELEHARA
RGARIYAELVGFGMSGDAFHMTAPPEDGAGAARCMKNALRDAGLDPRQVDYINAHGTSTPAG
DIAEIAAVKSVFGEHAHALSMSSTKSMTGHLLGAAGAVEAIFSVLALRDQVAPPTINLDNPDEGC
DLDLVAHEAKPRKIDVALSNSFGFGGTNGTLVFRRFAD
Id: Protein (A,B), Ligand (C).

## 2.5 Instruments, equipment and software for experimental validation of predicted binders

| Name | Use | Manufacturer |
|---|---|---|
| 5mL HisTrap HP column. | Retard His-tagget proteins. | GE healthcare |
| HiLoad 26/600 Superdex 75 column. | Purify proteins by size exclusion. | SigmaAldrich |
| ÄKTA pure 25 | Load sample for purification, collect flowthrough and elute, rinse columns. | Cytiva |
| Creaoptix WAVEdelta (CX-403 – 244604) | Measure kinetics of small molecules by RAPID. | Malvan Panalytical |
| 4PCH-NTA chip | Sensor and matrix for detecting phase shift for binding of analyte. Inserted into WAVEdelta. | Malvan Panalytical |

| | | |
|---|---|---|
| 1:1 Direct kinetics software. | Convert the measured phase shift to the interpretable dissociation constand Kd. | Malvan Panalytical. |
| Triple Sitting Drop 96-well plates | Induce crystal formation. | TTP Labtech |
| Amicon Ultra 10K | Concnetration of protein sample. | Sigma Aldrich |
| Nanodrop One$^{C}$ | Measure protein concentration. | Thermo Scientific |
| Biomax beamline | Collect X-ray data of crystals. | MaxIV synchroton radiation facility Lund, Sweden. |
| AutoPROC(Vonrhein et al., 2011) | X-ray diffraction image processing. | Global Phasing |
| EDNA2 | X-ray diffraction image processing. | European Synchrotron Radiation Facility (ESRF) copyright. |
| DIMPLE (Winn MD, 2011) | Molecular replacement | CCP4 |
| Coot (Emsley et al., 2010) | Inspection and correction of structure. | P.Emsley, B.Lohkamp, W.G Scott, K.Cowtan. |
| Phenix.refine (Afonine et al., 2012) | Refinement and processing of structure. | Phenix |

## 2.6 Buffers and growth medium of protein expression and purification

**Autoinduction medium: (1L)**
1.0% (w/v)Tryptone (T7293 sigma-aldrich)
1.0% (w/v) yeast extract (BP1422-2 Fisher Bioreagents)
30 µg/mL Chloramphenicol
50 µg/mL Kanamycin
0.5% Glycerol (G5516 Sigma-Aldrich)
0.05% Glucose (G7021 Sigma-Aldrich)
0.2% α-lactose (G17814 Sigma-Aldrich)
50 mM $Na_2HPO_4$
50 mM $KH_2PO_4$
25 mM $(NH_4)_2SO_4$
1 mM $MgSO_4$

**LB medium: (1L)**
1.0% (w/v)Tryptone (T7293 sigma-aldrich)
1.0% (w/v) yeast extract (BP1422-2 Fisher Bioreagents)
0.5% (w/v) NaCl

**Bacterial Plasmid stock and Avi + His tagged PaFabF C164A plasmid**
E.coli BL21(DE3)pLysS
Plasmid: FabF C164A AviTag pET-28a-TEV (GenScript)

**Lysis buffer**
50 mM $Na_2HPO_4$, 300 mM NaCl, 20 mM imidazole, 10% glycerol, 0.5 mM DTT, pH 7.6, 1 tablet of EDTA-free protease inhibitor (Roche, cat. No.11873580001), 20 µl of DNase I (Sigma, Cat.No.D5052).

**Immobilized metal affinity chromatography buffer**
Buffer A: 50 mM $Na_2HPO_4$, 300 mM NaCl, 20 mM imidazole, 10% glycerol, 0.5 mM DTT, pH 7.6)
Buffer B: 50 mM $Na_2HPO_4$, 300 mM NaCl, 500 mM imidazole, 10% glycerol, 0.5 mM DTT, pH 7.6

**Size exclusion chromatography buffer**
Buffer C: 50 mM Na2HPO4, 150 mM NaCl, 10% glycerol, 1 mM DTT, pH 7.4

**TEV cleavage**
TEV protease supplied by Khanh Kim Dao.

## 2.7: GCI buffers and solutions

Buffer B: 1x PBS, 1mM DTT (D0632 Sigma-Aldrich), 0.004%TritonX, 5% (v/v) Dimethyl sylphoxide (DMSO Sigma-Aldrich 151874)
Borate buffer: pH 9, 1 M NaCl (1066690010Sigma-Adrich)
EDTA: ( E5134 Sigma-Aldrich)

## 2.8: X-ray crystallography buffers, conditions and structure

Protein Buffer: 50 mM Na2HPO4, 150 mM NaCl, pH 7.4
Conditions: Formiate (09735 Sigma-Aldrich), PEG3350 (202444 Sigma-Aldrich)
Molecular replacement structure: 8CN8 (Georgiou, 2023a)

# 3. Methods

## 3.1 Assembly of the virtual screening library

A library of 1368 compounds referred to as ENAMINE-EOS set was created from a SMARTS-based substructure search for compounds listed by the chemical company ENAMINE. The used SMARTS pattern (Figure3.1) describe the key scaffolds of the top three hits identified in recent fragment scree of EU-OPENSCREEN fragment library. (Jalencas et al., 2024).

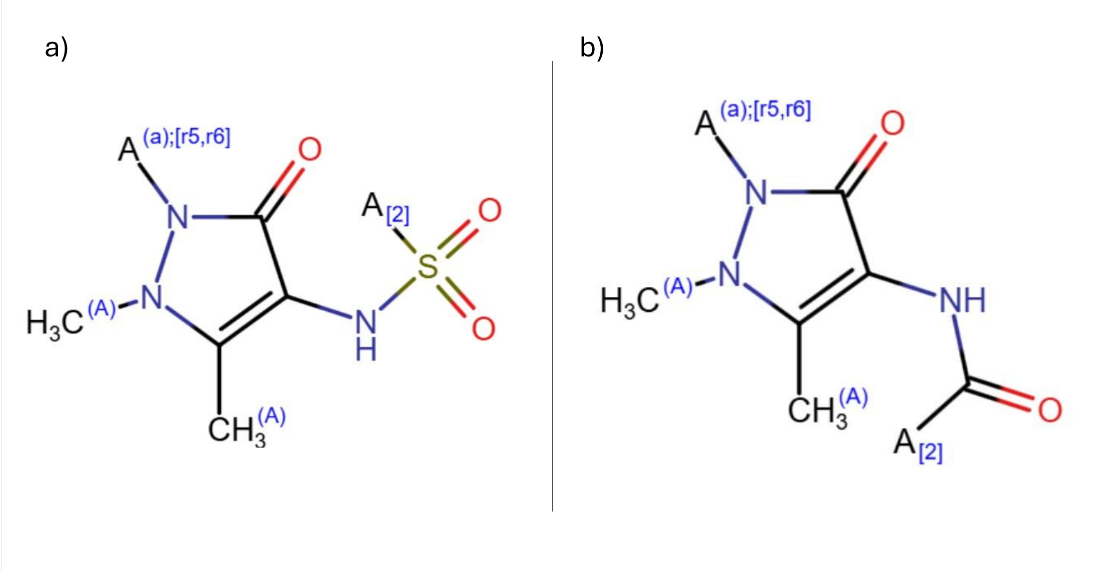



***Figure 3.1:*** *Chemical scaffold criteria used to select compounds provided by ENAMINE for a PaFabF C164A focused screening library. (A)$CH_3$ defines attachment points for substituents whereas the connecting atoms neds a carbon atom. The SMARTS pattern "(a);[r5,r6]" defines aromatic atoms as part of a 5 or 6 membered ring and $A_{[2]}$ allows any atoms as substituents. (a) represents the SMARTS used for the sulphonamide scaffold. b) represents the SMARTS used for the amide scaffold (Materials 2.1).*

## 3.2 Ligand preparation

From the SMILES provided by ENAMINE, inhouse scripts (Materials 2.2) based on RDKit prepared the charged states and tautomeric forms at neutral pH of the compounds. Conformers and possible stereoisomers, were generated using LigPrep from Schrödinger without changing the precalculated protomers and tautomer's (materials 2.2). This resulted in 2254 prepared small molecules.

## 3.3 Molecular Docking

All models except Boltz-2 and iScore required a 3D docking pose of the ligand inside the PaFabF C164A protein malonyl binding pocket in order to predict binding affinity. Both the FlexX(Matthias Rarey1*, 1996) and Glide(Halgren et al., 2004) docking engines were used to generate poses (Materials 2.3). For both docking engines, the PDB model 8R0I was used as a template.

For FlexX, the binding site was defined by using the ligand in 8R0I as reference ligand. The docking gris was automatically created around this ligand. The prepared ligands in SMILES format were docked using FlexX default settings (10 maximum poses, Standard clash tolerance, allowing ring conformations, not allowing stereo centre flipping. A redocking of the 8R0I, 8R1V and 8PJ0 ligands with the same settings gave a respective RMSD of 0.52Å, 0.73Å and 0.6Å compared to the crystal poses. RMSD was calculated with the obrms Obabel tool (Materials 2.3). The docking was run on Ubuntu 24.04 4LTS and generated 19830 poses.

Docking using Glide required a prepared protein. The biological assembly 1 of 8R0I was chosen due to a lower B-factor compared to biological assembly 2. First, the protein PDB file was prepared by the "Protein Preparation Workflow" in Maestro. The preparation of the protein was performed at a pH 7.4 and included the addition of missing hydrogens, optimisation of H-bond assignments, energy minimization and removal of waters far from HET groups. Four  water molecules in the binding site were removed due to not clashing with the larger 8R1V ligand (Figure3.2). Docking of the prepared ENAMINE-EOS set in mol format was done by a docking grid created with default settings around the 8R0I ligand. For docking, Thr 306 and Thr 308 were set as rotatable. Redocking of the ligands of the three PDB structures 8R0I, 8R1V and 8PJ0 gave a respective RMSD of 0.34Å, 0.47Å and 0.88Å compared to the crystal poses.

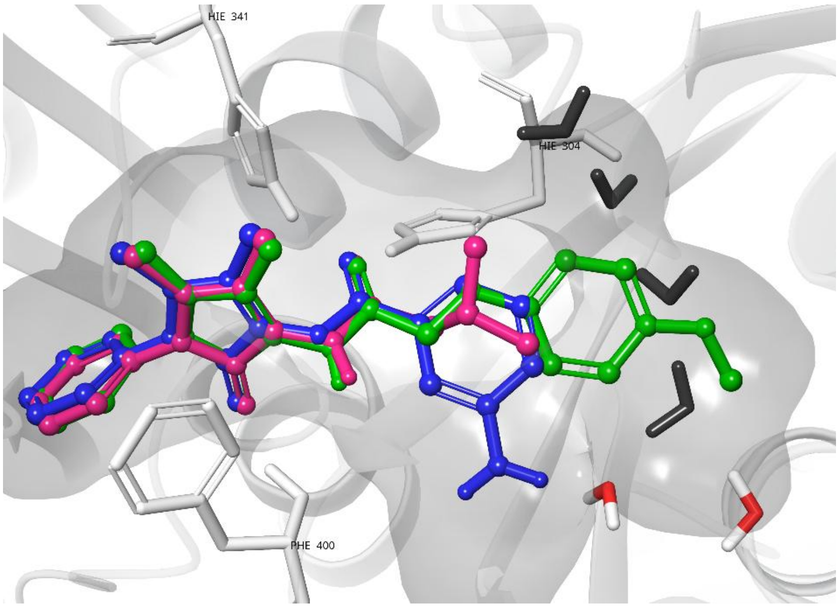



***Figure 3.2:*** *Overlay of the ligands from PDB structures 8R0I (yellow), 8R1V (green), and 8PJ0 (blue) within the 8R0I binding pocket. Black spheres represent water molecules removed due to clashes with the larger 8R1V ligand. The two water molecules which were kept are coloured as red and white. Figure was made in Maestro (2026-2).*

## 3.3 Rescoring using ML-based scoring functions

How each ML-based scoring function was used in order to predict binding affinity of small molecules in the ENAMINE-EOS set towards *Pa*FabF C164, is stated below. See Materials 2.4 for dependencies and computer specifications used to run each ML-based scoring function.

Kdeep: Scoring of docked poses in sdf format against FlexX and Maestro prepared 8R0I pdb structure was done using the PlayMolecule.ai platform. Calculations were cloud based (Jimenez et al., 2018). The amount of poses that could be scored against a protein exceeded the total number of poses. Thus the scoring was done in batches of up to 3333 poses.

vScreenML v2.0: Scoring of the ENAMINE-EOS set against the prepared 8R0I required 4 steps: Ligand parametrization where pdb and param files were generated from ligands in MOL2 format. Protein-ligand complex minimization by protein pdb file and both ligand pdb and param files were inputted to generate a protein-ligand complex in pdb format. Feature calculation of protein-ligand pdb file and ligand param file giving a minimized complex feature CSV file. A prediction score of the complex CSV file were then scored giving a prediction CSV file containing a value between 1 to 0 based on likelihood of binding (Andrianov et al., 2024). The predictions were ranked based on their likelihood of being a binder.

Ak-Score2: Prior to affinity prediction, the ligands were converted by AK-Score2 script "convert_mol2pdb.py" from MOL2 to pdb format. The inference command used for scoring was: python akscore2_run.py -r protein.pdb -l ligand.pdb -s akscore2_docks -o results.csv --device cpu.

iScore: iScore does not require a pose as it predicts binding affinity (pKd) based on a set of descriptors calculated from SMILES of the ligand and binding pocket descriptors of the target protein. PaFabF C164A pocket descriptors was created by dpocket of fpocket dependency with command: dpocket -f input.txt -o output. Ligand descriptors was generated by Descriptors.py(Landrum, 2024). Inference was performed by scaling the prepared iScore feature matrix with the provided pretrained MinMaxScaler, applying the pretrained DNN, Random Forest, and XGBoost ensemble models, averaging the predictions within each ensemble, and using the three averaged base-model predictions as input to the pretrained Hybrid model. The Hybrid output was reported as the final iScore prediction.

OnionNet22: OnionNet22 required each ligand pose in complex with the agonist protein as a PDB format. Scoring was run by the inference command: predict.py -rec_fpath protein.pdb -lig_fpath ligand.pdb -shape 84,124,1 -scaler OnionNet2_path/models/train_scaler.scaler -model OnionNet2_path/models/62shell_saved-model.h5 -shells 62 -out-path output.csv The binding affinity was received in pKa scale (Zheng et al., 2019).

Boltz-22: For screening, Boltz 2 required protein sequence and ligand SMILES as input. Boltz-2 predicted binding affinity and co-folding of the protein-ligand complex by the command boltz predict file.yaml --out_dir output_dir --use_msa_server --output_format pdb. The binding affinity was received by an IC50-like value (Passaro et al., 2025).

## 3.5 Selection of 5 compounds from top 1%

From each screening, 5 compounds from each scoring functions top 1% was selected for prospective validation. Selection was primarily based on the top 5 ranked compounds. However, compounds with characteristics problematic for affinity measurements such as poor solubility and potential reactive groups, were excluded in favour of more suitable compounds within the top 1% rank. Compounds that overlapped between the top1% ranks, were prioritised.

## 3.6 Analysis of virtual screening

Heavy atoms per molecule and spearman correlation between rankings obtained using the different scoring function were calculated using Python scripts making with SciPy and Rdkit dependencies. The same dependencies was used to calculate the median score obtained with the different methods. Heatmap plots were generated using

## 3.7 Expression of PaFabF C164A

800µl of bacterial glycerol stock transformed with PaFabF C164A plasmid (Materials 2.6) was inoculated with 100mL of LB medium (Materials 2.6. After incubating the solution overnight at 37°C with shaking at 220rpm, 10mL of preculture was further inoculated with 1L of autoinduction medium Materials 2.6) and further incubated at 20°C at 200rpm shaking for 48 hours. After incubation, the culture was placed on ice for 10 minutes. Cells were then harvested from the culture-solution by a centrifugation at 5500g for 45 minutes at 4°C. The cell-pellet formed from centrifugation were washed with 0.9%NaCl and further centrifuged at 2600g for 45 minutes. The centrifuges pellet was then frozen in liquid nitrogen and stored at -20 until purification.

### 3.7.1 Purification of PaFabF C164A

To disrupt cells, the frozen pellets were resuspended in 30mL of lysis buffer (Materials 2.6) by stirring for 20 minutes at 4°C. The resuspended cells were then subjected to sonification for 10 minutes with 6s pulses. Insoluble proteins and cell debris were pelleted by centrifugation at 16000g for 40 min followed by syringe filtration through a Whatman Puradisc 0.45 µm filter.

The cell lysate was first purified by immobilized metal affinity chromatography (IMAC) using a 5mL HisTrap at 20°C with an Akta Pure 25 (Materials 2.6). The HisTrap HP column (GE Healthcare) was equilibrated with Buffer A (Materials 2.6). The cell lysate was then loaded onto the HisTrap column. To remove unwanted content, the column was washed with Buffer A until a stable OD280 baseline on Aktar was reached. The protein was eluted from the column by adding a 0-100% gradient of Buffer B(Materials 2.6). The elute fractions containing the protein were concentrated by ultrafiltration using Amicon UltraImage 10K (Milipore).

The elute from HisTrap purification was further purified by size exclusion chromatography (SEC) using HiLoad 26/600 Superdex 75 pg column. The HisTrap purified sample was centrifuged at 13500g for 10 minutes before loading onto the SEC column equilibrated with gel filtration buffer Buffer C (Materials 2.6). The column was washed by 1.5 column volumes (CV) of Buffer C and the elute was collected in fractions. The Fractionated flow through was concentrated using Amicon Ultramage 10K to a concentration of 10.9mg/ml by nanodrop. The concentration was measured by Nanodrop using molecular weight of 48.155 and molecular extinction coefficient $\epsilon$ 28670.

### 3.7.2 Cleavage of His tag by TEV cleavage and reverse HisTrap

His-tagged PaFabF C164A was incubated with TEV protease at a ratio of 1:10 (TEV:protein) overnight at 4C with shaking. Solution was then loaded onto 5mL HisTrap column (Materials 2.5) on Akta pure 25 equilibrated with Buffer C (Materials 2.6). His-tag cleaved protein was collected in flowthrough fractions and concentrated using Amicon Ultramage 10K.

## 3.8 GCI

The first screening was done using a 4PCH-NTA chip with a Ni-NTA surface. The chip was inserted into Malvern Panalytical Creoptix WAVEdelta (CX-403 – 244604) and measurements were carried out at 25°C. The chip was first purged with degassed MilliQ water before conditioning with 0.1M borate buffer (pH9, 1M NaCl) to hydrate and expand the chip sensor matrix. To remove possible metal ions, the chip was then conditioned with 0.25M EDTA (pH 8). Both solutions had an injection duration of 180s and dissociation duration of 60s. All four channels of the chip was then injected with degassed MilliQ water at an injection duration of 60s and dissociation duration of 60s. This was repeated 3x and followed by three injections of 60s association and 60s dissociation by running buffer at a flow rate of 10µl/min.

Prior to immobilising the chip sensor matrix with $NiCl_2$, all 4 channels were rinsed twice with running buffer B at an injection duration of 60s and dissociation duration of 60s at a flow rate of 10µL/min. $Ni^{2+}$ was then immobilised by injecting 0.5mM $NiCl_2$ for 120s and dissociate for a duration of 60s at a follow rate of 10µl/min. The chip channels were then rinsed with running buffer B twice by an injection of 60s and dissociation of 60s with a flow rate of 10µl/min.

After a 2x rinse of running buffer B with an injection and dissociation duration of both 60s with flowrate of 10µL/min the His-tagged protein PaFabF C164A was captured on channel 2 at a density of 6191.0 pg/$mm^2$ with an injection time of 420s and dissociation time of 60s.

Purchased ligands and PTM acting as a positive control was then measured by waveRAPID kinetics with adjusted settings for weak binders at a total association of 15s and a dissociation duration of 100s at a flowrate of 200 µL/min for each compound. The compounds had a concentration of 200µM in solution with running buffer B and 5% DMSO. PTM diluted with a concentration of 10µM in 5% DMSO in running buffer B. PTM was injected three times with an interval of 10 compounds.

## 3.8.1 GCI kinetic fit quality assessment

The 1:1 Direct Kinetics program of Malvan WaveRAPID software was used to calculate the dissociation constand Kd of the interaction between the purchased compounds from ENAMINE and His tagged PaFabF C164A. For each Kd calculation, Direct Kinetics provided Rmax, Kd error (%) and Chi2. Classification of the screened compounds as binders, were based on cut off values for Kd error (%), Rmax and visual inspection of sensorgrams used in previous WaveRAPID-based screening studies (Pal et al., 2024) To be classified as a binder, the compound were required to exhibit a fitted Rmax value > $2pg/mm^2$, dissociation rate (Kd) error < 60 % and sensorgrams displaying a concentration dependent binding.

Furthermore, the binders were divided into two categories based on accuracy of the calculated Kd. GCI use the same Ordinary Differential Equations (ODEs) as traditional kinetics (Kartal et al., 2021) Reported quality standards for traditional kinetics including SPR (Belcher et al., 2024) state Chi2 value < 10% as high kinetic fit confidence (Luo, 2013). Furthermore, the Kd calculation was determined to be inaccurate if the calculated Rmax was higher than the theoretical Rmax (Watts, 2022).

## 3.9 X-ray crystallography - Co-crystallisation

All compounds with a calculated Kd below 100µM and an Rmax above $2pg/mm^2$ from the GCI affinity measurements were selected for X-ray crystallography Figure 4.3.1, 4.3.2, 4.3.3. In case the decided cut-off values were inaccurate, the selected compounds included four binders with inaccurate Kd prediction and one compound described as a non-binder. In addition, a compound not included in the single point GCI screening was also subjected to co-crystallisation. The 8 The compounds were diluted in 3% DMSO with protein buffer (Materials 2.8) and were incubated with 10mg/ml His cleaved PaFabF C164A at 4°C overnight. The 7 compounds with final concentration included: SN-1221(7.5 mM), SN-2288 (4.9 mM), SN-1234 (7.5 mM), SN-1877(7.5 mM), SN-0709 ( 3.8 mM), SN-0385 (7.5 mM), SN-0163 (3.5mM) and SN-1442 (3.8mM). Each protein–ligand complex was subjected to six different reservoir conditions on a Triple Sitting Drop 96 vapor diffusion plate different reservoir **Table3.1**. For each condition the volume was 40µl. The plate was stored at 20°C to induce complex crystallisation.

| 0.24M $NH_4$ Formate | | | 0.24M $NH_4$ Formate | | |
|---|---|---|---|---|---|
| 18%PEG3350 | 21%PEG3350 | 27%PEG3350 | 18%PEG3350 | 21%PEG3350 | 27%PEG3350 |

**Table3.1**: The six different conditions each co-crystal was subjected to on a Triple Sitting Drop 96 vapor diffusion plate.

### 3.9.1 X-ray data acquisition, processing and refinement

Crystals were fished, cryoprotected with 20% glycerol and frozen in liquid nitrogen. X-ray data were collected at 100 K at the Biomax beamline of the MaxIV synchrotron radiation facility in Lund Sweden. X-ray diffraction images were processed using the auto process pipelines provided by the beamline (autoproc, EDNA2). Dimple from the CCP4i2 suite was used for molecular replacement using the PDB ID: 8CN8 as input model. Modelled structures were manually inspected and corrected using Coot while phenix.refine from the phenix suite was used for the refinement. Data collection, processing and refinement statistics are listed in Table2 of the appendix.

## Work done by others

Marie Thiemann expressed and purified the His cleaved PaFabF C164A used for co crystallisation of PaFabF C164A and ligand complexes. Marie Thiemann also fished and refined the SN-1221 and SN-0709 crystal complex from X-ray diffraction data. Iris Rami performed screening of the ENAMINE-EOS set by the three scoring functions iScore, Boltz-2 and OnionNet2. The manual selection was a collective effort by Sigurd S.Nøklebye, PhD candidate Alberto Memè and Prof. Dr. Ruth Brenk.

# Results

## 4.1 Rescoring of ENAMINE-EOS set

In this thesis, I aimed at evaluating six different ML-based scoring functions by applying them in the prediction of new ligands for PaFabF C164A. As a virtual screening database I used a focused library containing analogues of previously discovered hits found in a screening campaign with the EU-OPENSCREEN initiative towards the target protein PaFabF C164A (Jalencas et al., 2024). This database referred to as the Enamine-EOS set was generated by a substructure search on the Enamine compound catalogue using both a dimethyl-phenyl-pyrazolone scaffold with a secondary amide, and a dimethyl-phenyl-pyrazolone with a sulphonamide. The final Enamine-EOS set contained 1368 small molecules. From the SMILES listed for each compound in the Enamine-EOS set, different ligand states were generated by Brenk-lab scripts and Maestro ligprep resulting in a total of 2254 SMILES.

Four out of the six scoring functions required ligand poses against the 3D structure of the target binding pocket in order to predict binding. These scoring functions included AK-Score2, OnionNet2, KDeep and vScreenMLv2.0. The prepared ligands were docked against the 8R0I pdb structure using GLIDE and FlexX resulting in a total of 28.417 docked poses. For the generation of poses by Glide, the 8R0I protein was prepared in Maestro (version). For the two scoring function iScore and Boltz-2 which did not require a pose. iScore used descriptors made from the ligand smiles and protein pocket to predict binding. Boltz-2 generated its own protein-ligand complex based on ligand SMILES and protein sequence in order to predict binding.

The rank provided by each scoring function provided an opportunity to validate the predictions by later screening the top ranked compounds experimentally. During the screening, three scoring functions failed to score all 1368 compounds of the ENAMINE-EOS set. vScreenMLv2.0 failed to score 2 compounds, Boltz-2 failed to score 2 compounds and AK-Score2 failed to score one compound. To obtain a single ranking entry per compound, the ligand state and corresponding pose with the highest prediction value were selected to represent its corresponding ENAMINE listed compound. The final compound ranking consisting of 1368 small molecules.

### 4.1.1 Final selection from rescoring of ENAMINE-EOS set

In total, 30 compounds were selected for in vitro screening. As shown in Table4.1.6 out of 7 selections shared at least one overlap with another selection. KDeep shared the greatest overlap, containing compounds also chosen by the iScore, OnionNet2 and Manual selection. The Boltz-2 selection shared no overlap with the ones from the other models.

| AK-Score2 | Boltz-2 | Iscore | KDeep | OnionNet2 | VScreenMLv2.0 | Manual selection |
|---|---|---|---|---|---|---|
| SN-0913 | SN-0163 | SN-2317 | SN-1221 | SN-0366 | SN-0385 | SN-0729 |
| SN-0528 | SN-0216 | SN-1254 | SN-0516 | SN-1242 | SN-0368 | SN-0314 |
| SN-2437 | SN-0953 | SN-2241 | SN-1254 | SN-1442 | SN-0709 | SN-1234 |
| SN-1877 | SN-0738 | SN-1877 | SN-1242 | SN-0746 | SN-1251 | SN-1221 |
| SN-0385 | SN-0156 | SN-2166 | SN-2288 | SN-0846 | SN-0109 | SN-0073 |

***Table 4.1: The 5 selected compounds by each scoring function.*** *Each row displays the small molecules selected from the top 1% rank from each scoring function with the exception of the Manual selection which was selected by visual inspection of GLIDE generated poses. Instances with overlap between selected compounds is highlighted with matching colours.*

### 4.1.2 Analysis on agreement of models in rescoring

Each scoring functions goal is to provide the most accurate prediction of small molecules binding strength towards their proposed target. Each of the six scoring function employ a different method to predict a small molecules binding strength. A comparison of the rank provided by each scoring function was done in order to assess whether they agreed despite their difference in ML architecture and training.

The three hits found in the screening campaign against PaFabF C164A using EFSL—more—library referred to as the 1st gen EOS series, all shared the same key interactions as depicted in Figure1.1 with their dimethyl-phenyl-pyrazolone motif. The assumption when assembling the ENAMINE-EOS dataset was that these molecules would follow the same binding mode. The analysis shown in Figure4.1.2 was done by visual inspection of the top scored pose for each small molecule in the top1% rank to validate whether the scoring functions were able to retain known dimethyl-phenyl-pyrazolone interactions. Figure4.1.2 show that all of the top 1% ranked small molecules by vScreenMLv2.0 displayed this interaction. In contrast, between 30% to 50% of the top 1% ranked small molecules by KDeep, AK-Score2 and OnionNet2 retained the dimethyl-phenyl-pyrazolone motif interaction. The model Boltz-2 did not rank any compounds with this key interaction in their top 1% rank.

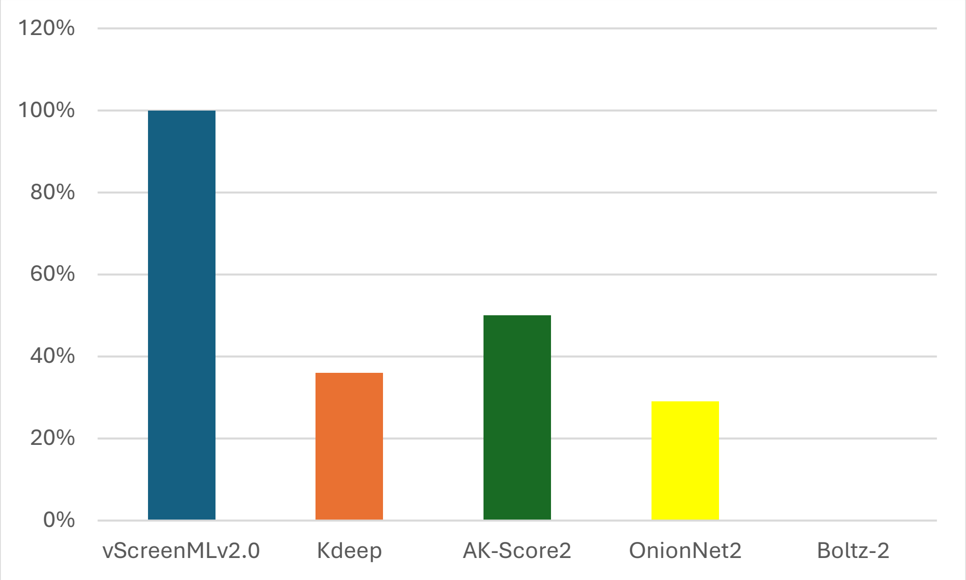



***Figure4.1.2: Binding mode distribution of top 1% ranked small molecule poses.*** *Distribution of poses which follow the same validated binding mode of the dimethyl-phenyl-pyrazolone from each models top 1% rank.*

A method to asses the level of agreement between the six scoring functions in their ranking of the ENAMINE-EOS set, was plotting the heavy atom count for every small molecule against their predicted rank. The analysis could provide insights to whether they ranked compounds according to different principles. It could indicate whether the scoring functions tended to favour larger molecules, a bias commonly observed for classical scoring functions (Kitchen et al., 2004).

The scatter plots displayed in Figure4.1.3 show no general agreement between all the six ML-based scoring functions in terms of how they ranked molecules based on their heavy atom count. The rankings produced by AK-Score2, OnionNet2, iScore, and KDeep exhibited statistically significant relationships of $p < 0.05$ between heavy atom count and ranking position, with compounds containing a higher number of heavy atoms generally receiving more favourable rankings. Ranking position accounted for between 4.8% and 41.4% of the variation in heavy atom count according to the respective calculated $R^2$ of 0.048, 0.152, 0.414 and 0.333. In contrast, the rank produced by Boltz-2 exhibited a statistically significant trend where an increase in heavy atoms per compound generally led to a lower rank. For Boltz-2, 16.5% of the variation in heavy atom count were accounted for by the ranking position according to the listed $R^2$ of 0.165. For the model vScreenMLv2.0, no statistical significant trend $p > 0.05$ between the rank achieved by each compound and its heavy atom count, was observed.

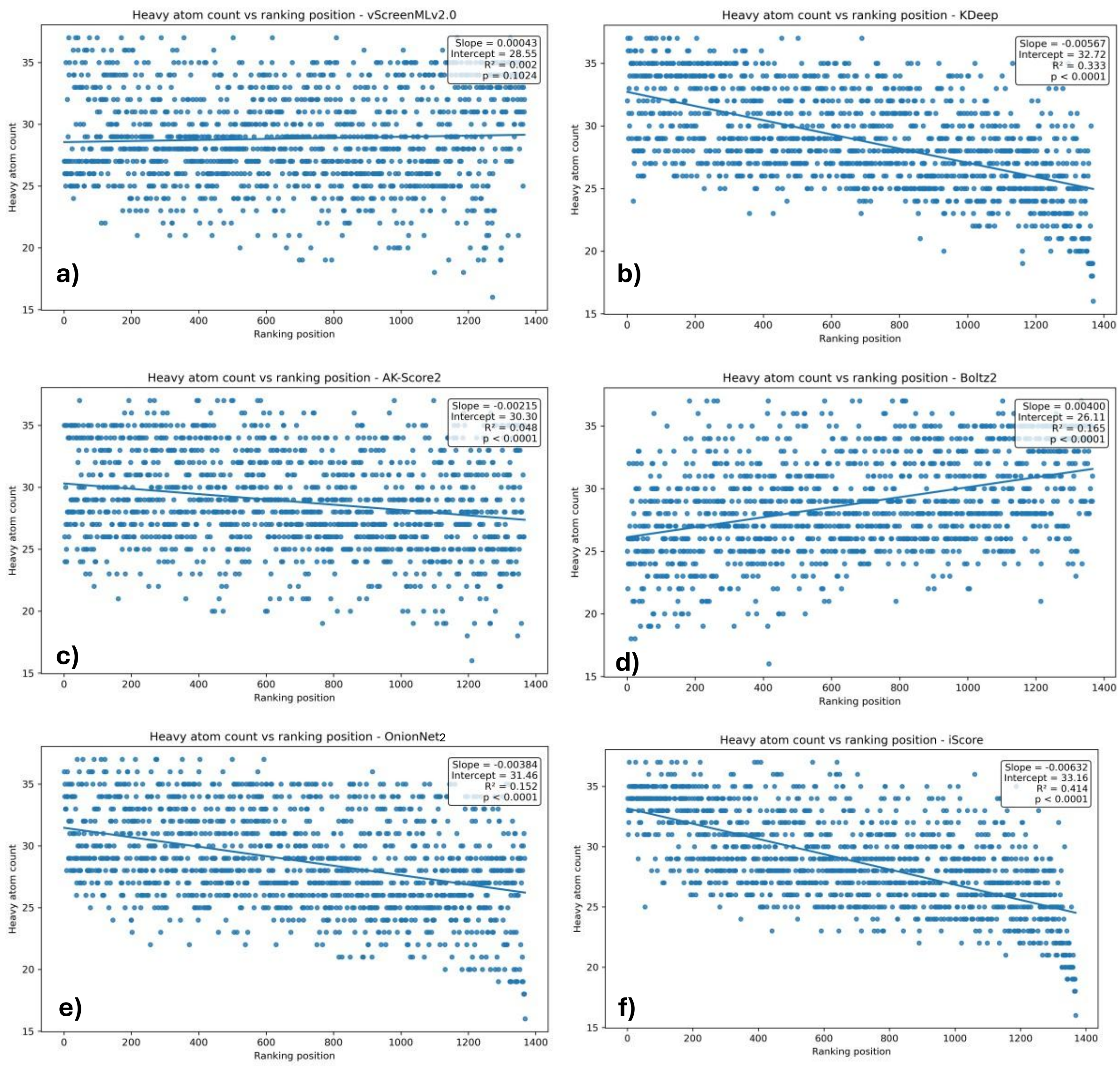



**Figure4.1.3: Heavy atom count vs predicted rank.** *Relationship between heavy atom count for the compounds in the ENAMINE-EOS set and their predicted ranking position displayed as a scatter plot for the **a)** vScreenMLv2.0 rank, **b**) KDeep rank, **c)** AK-Score2, **d)** Boltz-2, **e)** OnionNet2 and **f)** iScore rank. Position 1 correlate to the predicted strongest binder. The fitted linear regression line is shown in each plot. The regression slope, coefficient of determination ($R^2$), and p-value are reported in the upper-right corner of each plot. Scatter plots and linear regression models were generated in python using Matplotlib.*

If two ENAMINE-EOS set was ranked by chance, there would be a 5% overlap in the top 5% rank. As shown in Figure4.1.4a, many of the six scoring functions achieved a pairwise similarity close to 5% overlap in the their top5% ENAMINE-EOS rank towards. Generally most scoring functions only shared a pairwise overlap greater than 5-6% with only two other scoring functions. Boltz-2 was the exception as it exhibited overlap percentages that were generally close to or below those expected from a random selection while showing little to no overlap with several of the other rankings. Boltz-2 shared 0% overlap between the both the rank produced by iScore and KDeep. On the other hand, KDeep and iScore shared the highest percentage of overlap at 23% between their top5% ENAMINE-EOS rank. iScore and AK-Score2 shared the second highest overlap at 22% top5% rank similarity. As shown in Figure4.1.4b, within the top 25%ranks KDeep and iScore also shared the highest similarity of 55% overlap. The AK-Score2 and iScore rank had the second highest overlap at 35% within their top 25% ranks. The overlap pattern among the scoring functions top 25% rank relative to the overlap expected by chance, was consistent with the observations of Figure4.1.4a.

A method to asses the level of agreement would be to check how other scoring functions rank the top 1%. The median percentile placement of the top1% in other ranks could give an indication for the level of agreement. The heatmap displayed in Figure4.1.4c show a large spread ranging from 8.5% to 94.2% of where each scoring functions top 1% was ranked in the other scoring functions global rank. The heatmap show that the highest median rank placement is found for iScore's top 1% achieving a median rank of 8.9% in the global KDeep rank and a 8.5% median rank placement in the OnionNet2s global rank. KDeep and OnionNet2s top1% only achieve a median ranking of 18.4% and 54.9% respectively in iScores full rank. Boltz-2 generally achieved the lowest median percentile rank of its top 1% ranked compounds. The small molecules in Boltz-2 top1%, had a median rank placement of 94.2% in iScores global rank.

A spearman rank correlation was done between the global ranks of all six scoring functions to further investigate whether the models are fundamentally similar, or if they rank compounds according to different principles. The Spearman rank correlation heatmap displayed in Figure4.1.4.d demonstrated an overall weak agreement between the global ranks produced by the six scoring functions on the ENAMINE-EOS set. With a $p=0.58$, the rank produced by iScore and KDeep had the highest correlation between the models. In contrast, the lowest correlation was found between the rank produce by iScore and Boltz-2 with a $p=-0.48$. Boltz-2 generally achieved the lowest correlation between the ranks of all of the scoring functions.

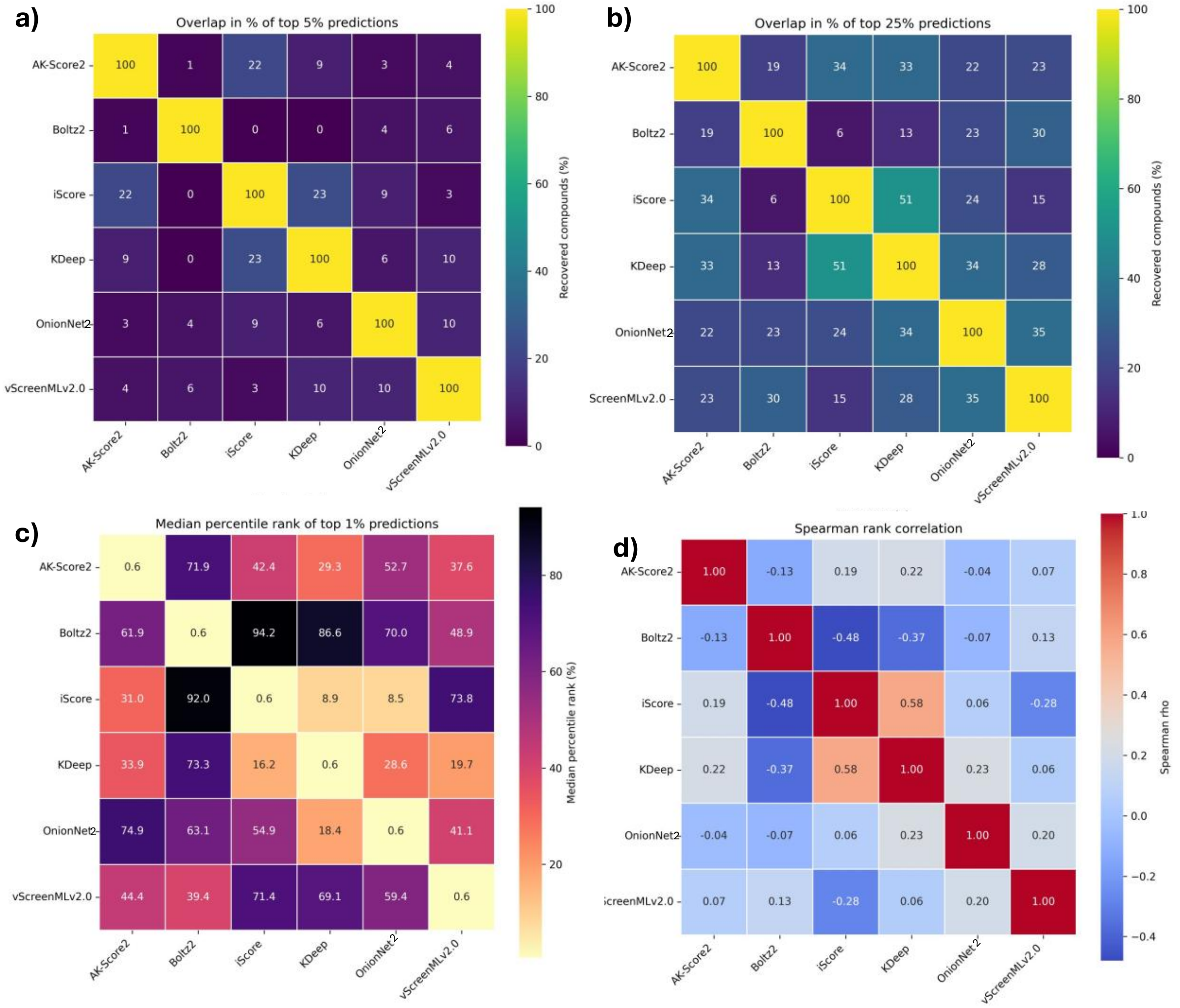



**Figure4.1.4: Pairwise ranking heatmaps.** ***a)*** *Heatmap of the overlap in percentage between the six ML-based scoring functions top 5% rank.* ***b)*** *Heatmap of the overlap in percentage between the six ML-based scoring functions top 25% rank.* ***c)*** *The median percentile rank of the six ML-based scoring functions top 1% rank in the global rank by the other five scoring functions.* ***d)*** *Pairwise Spearman correlation between the six scoring functions global rank. Heatmaps of a), b), c) and d) were made in python using matplotlib.*

## 4.2 Retrospective validation of rescoring of ENAMINE-EOS set

All six scoring functions have previously demonstrated promising performance in benchmarks based on experimentally validated data. To evaluate whether this performance translated to the PaFabF C164A malonyl binding pocket, the scoring functions were tested on their ability to identify the validated 1st gen EOS hits as binders.

Figure4.2 show that the six ML based scoring functions used in the virtual screening, differ in how they ranked the three confirmed low micromolar hits of the 1st generation EOS series. As shown in Figure4.2b, there is generally a relatively large spread between the three hits for each scoring function. Among the six scoring functions, Boltz-2 seems to have the highest median rank of the three compounds. In contrast, iScore has the lowest median rank for Hit 1, Hit 2 and Hit 3. The highest rank achieved was 8R1V predicted by KDeep which placed it around top 7%.

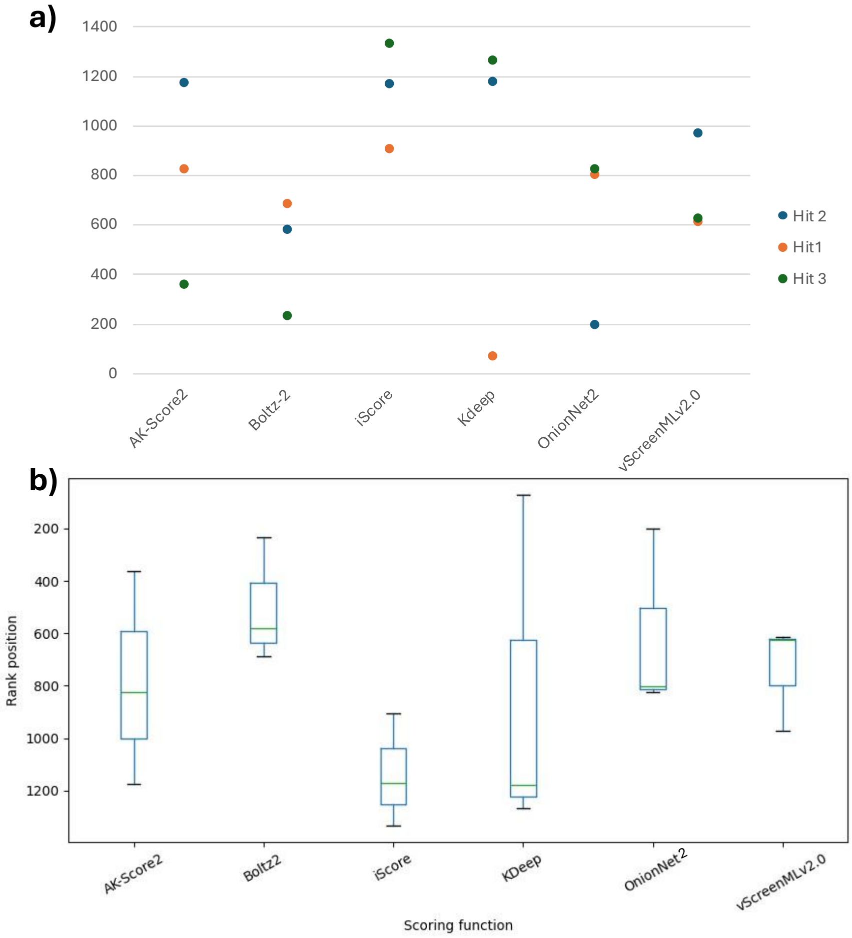


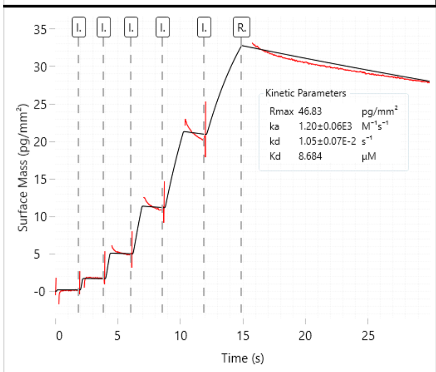
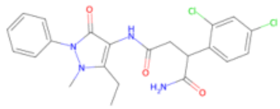
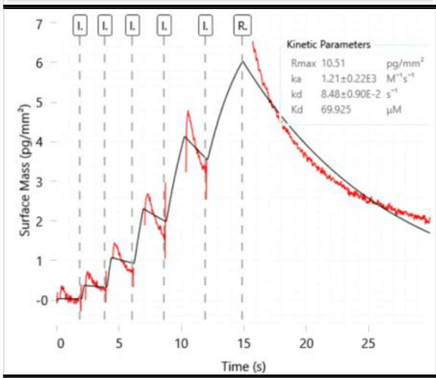
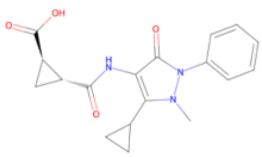
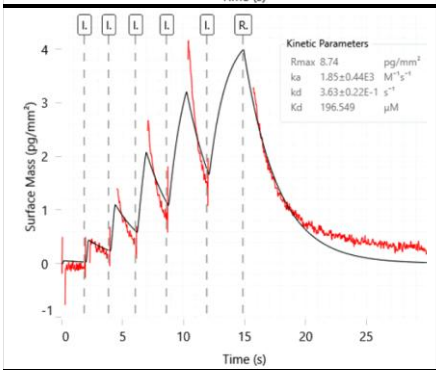
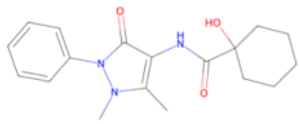

***Figure4.2: Re-scoring of 1st gen EOS hits. a)****The plot display the rank achieved by the three 1st gen EOS PaFabF C164A hitsin the virtual screening of the ENAMINE-EOS set by the six ML-based scoring functions AK-Score2, Boltz-2, iScore, KDeep, OnionNet2 and vScreenMLv2.0. A low rank corresponds to a predicted strong binder. b) Whisker plot summarising the distribution of the 1st gen EOS hits for each of the six ML-based scoring functions used in the virtual screening. Plots made in python using matplotlib and in excel.*

## 4.3 Prospective validation of virtual screening by affinity assay

A metric to determine the predicting power of a scoring function is its ability to enrich the top 1% rank with binders. Thus to evaluate the predicting power of the six scoring functions, 5 compounds from each of the top 1% was measured by GCI binding affinity assay. Measurements by Wave delta was chosen due to its stated high sensitivity in measuring both weak and strong binders by real time label free kinetics (Malvern-Panalytical, 2026). To further confirm binding, attempts were made to resolve PaFabF C164A-ligand complexes with X-ray crystallography.

## 4. 3.1 GCI affinity measurements

Eight compounds met the criteria for binders with accurate affinity calculations, as indicated by acceptable kinetic fit quality markers in figure4.3.1. SN-1221 was the small molecule with the highest affinity measured to a Kd of 8.7µM. It was selected by both the scoring function KDeep and through the  manual selection. With a measured Kd of 69.9µM, SN-0385 selected by both AK-Score2 and vScreenMLv2.0 had the second highest affinity towards PaFabF C164A. The other 5 compounds listed in Figure4.3.1 had a measured affinity above 100µM.

| Sensorgram. | Structure: | Model: | Kinetic fit markers: |
|---|---|---|---|
|  |  SN-1221 – ENAMINE-ID: Z5317952827 | KDeep, Manual-selection | Kd error: 7%<br>Chi2/Rmax: 8%<br>Rmax/th.Rmax: 140% |
|  |  SN-0385 – ENAMINE-ID: Z3510134710 | AK-Score2, vScreenMLv2.0 | Kd error: 11%<br>Chi2/Rmax: 8%<br>Rmax/th.Rmax: 40% |
|  |  SN-0528 – ENAMINE-ID: Z1386568501 | AK-Score2 | Kd error: 6%<br>Chi2/Rmax: 7%<br>Rmax/th.Rmax: 50% |

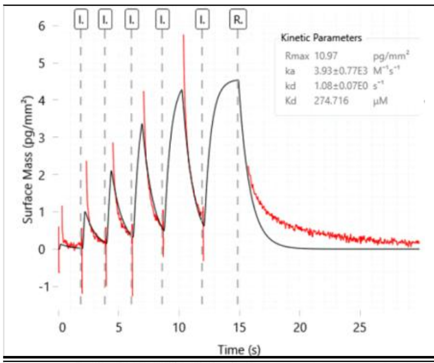
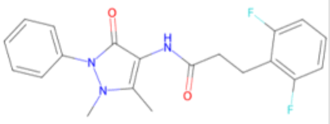
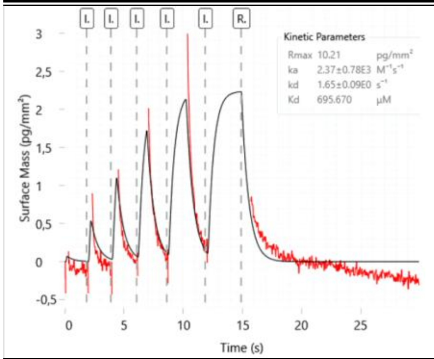
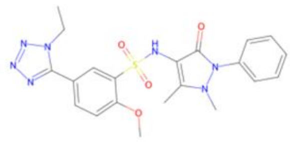
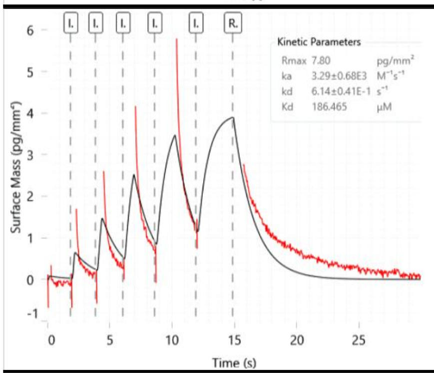
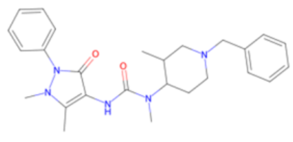
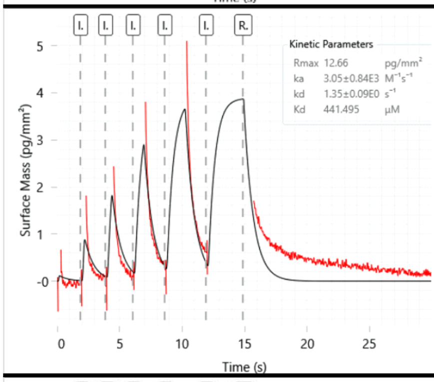
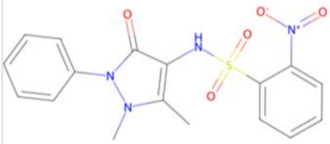
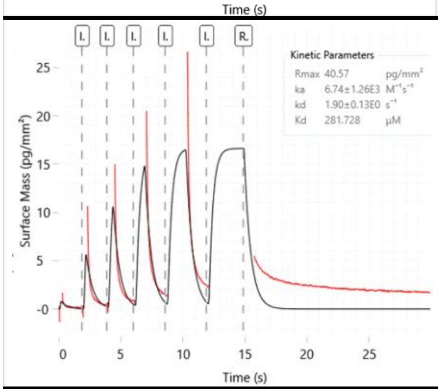
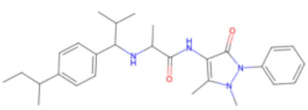
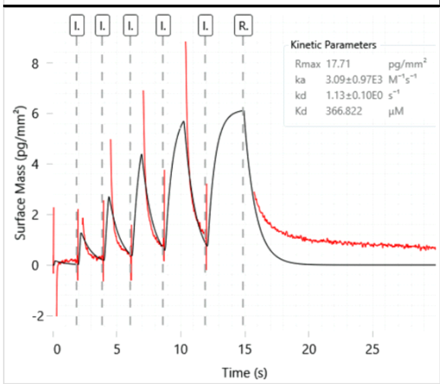
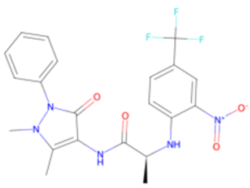
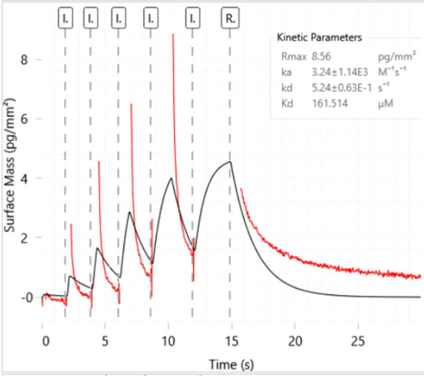
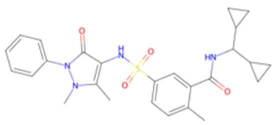
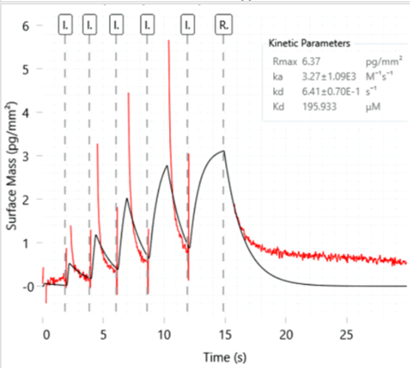
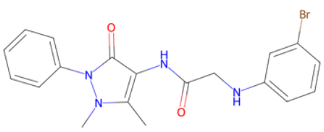
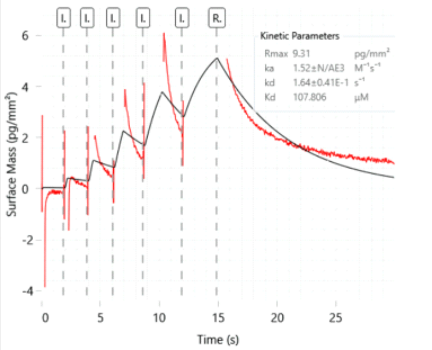
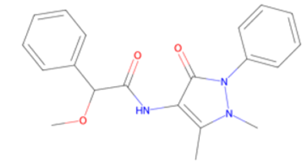
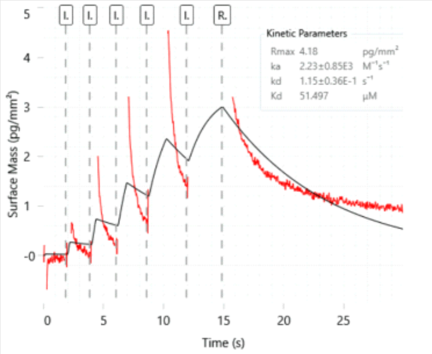
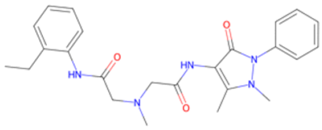
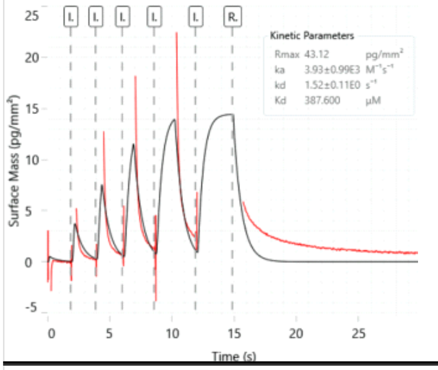
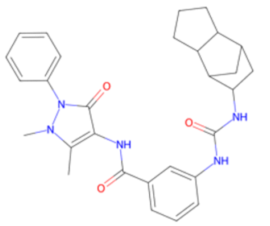
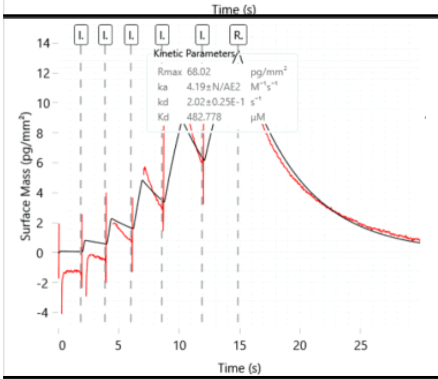
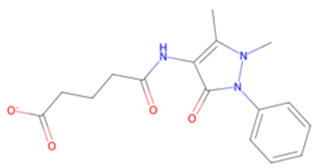
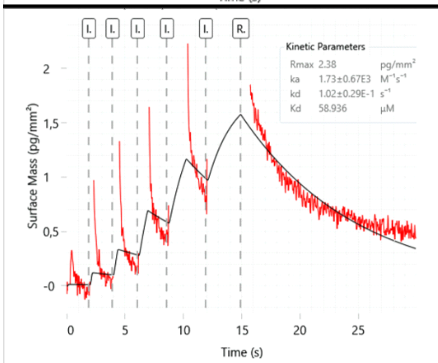
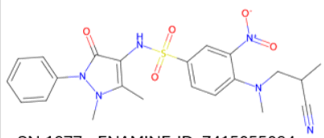
| | | | |
|---|---|---|---|
|  | <br>SN-0109 – ENAMINE-ID: Z382842398 | vScreenMLv2.0 | Kd error: 6%<br>Chi2/Rmax: 8%<br>Rmax/th.Rmax: 60% |
|  | <br>SN-2317 – ENAMINE-ID: Z56920091 | iScore | Kd error: 6%<br>Chi2/Rmax: 3%<br>Rmax/th.Rmax: 100% |
|  | <br>SN-0366 – ENAMINE-ID: Z1434685288 | OnionNet2 | Kd error: 7%<br>Chi2/Rmax:7%<br>Rmax/th.Rmax: 80% |
|  | <br>SN-2437 – ENAMINE-ID: Z45612762 | AK-Score2 | Kd error: 7%<br>Chi2/Rmax: 5%<br>Rmax/th.Rmax: 150% |
|  | <br>SN-0746 – ENAMINE-ID: Z106510796 | OnionNet2 | Kd error: 7%<br>Chi2/Rmax: 8%<br>Rmax/th.Rmax: 120% |

**Figure4.3.1: Confirmed binders from single point GCI screening.** Determination of dissociation constant Kd for small molecules fulfilling binder criteria of Rmax>2pg/mm², Kd<60% and with sensorgrams showing a clear concentration dependent binding response while achieving a direct kinetics fit accuracy indicator of Chi2/Rmax<10%.

As listed in Table 4.3.2, 12 compounds were categorised as binders with inaccurate affinity calculations. In general, inaccurate affinity estimates were associated with either poor direct kinetic fits above 10% or high Rmax/theoretical Rmax ratios, indicating over-stoichiometric binding conditions.

| Sensorgram. | Structure: | Model: | Kinetic fit markers: |
|---|---|---|---|
|  | <br>SN-0516 – ENAMINE-ID: Z57026795 | KDeep | Kd error: 8%<br>Chi2/Rmax: 12%<br>Rmax/th.Rmax: 50% |
|  | <br>SN-2166 – ENAMINE-ID: Z282950556 | iScore | Kd error: 12%<br>Chi2/Rmax: 15%<br>Rmax/th.Rmax: 20% |
|  | <br>SN-0913 – ENAMINE-ID: Z52832248 | AK-Score2 | Kd error: 11%<br>Chi2/Rmax: 27%<br>Rmax/th.Rmax: 20% |

| | | | |
|---|---|---|---|
|  |  SN-0156 – ENAMINE-ID: Z433859962 | Boltz-2 | Kd error: 25%<br>Chi2/Rmax: 19%<br>Rmax/th.Rmax: 50% |
|  |  SN-1234 – ENAMINE-ID: Z46437155 | Manual-selection | Kd error: 31%<br>Chi2/Rmax: 23%<br>Rmax/th.Rmax: 20% |
|  |  SN-1254 – ENAMINE-ID: Z411714436 | Manual-selection | Kd error: 7%<br>Chi2/Rmax: 11%<br>Rmax/th.Rmax: 190% |
|  |  SN-0953 – ENAMINE-ID: Z6655985202 | Boltz-2 | Kd error: 12%<br>Chi2/Rmax: 6%<br>Rmax/th.Rmax: 270% |
|  |  SN-1877 – ENAMINE-ID: Z415055084 | AK-Score2, iScore | Kd error: 28%<br>Chi2/Rmax: 17%<br>Rmax/th.Rmax: 10% |

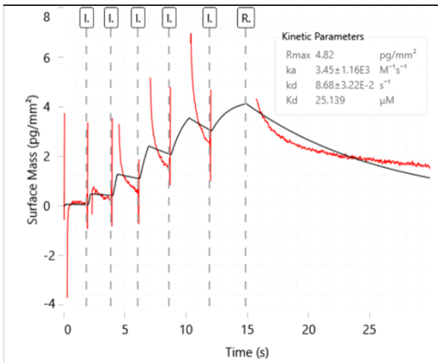
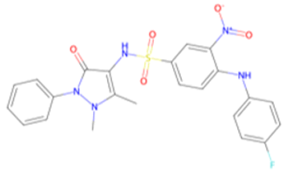
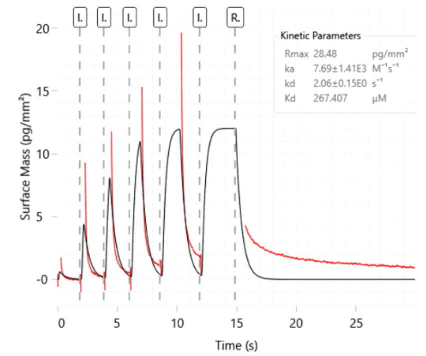
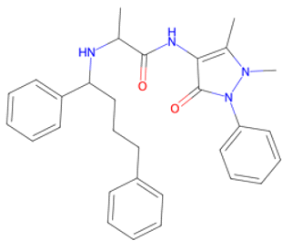
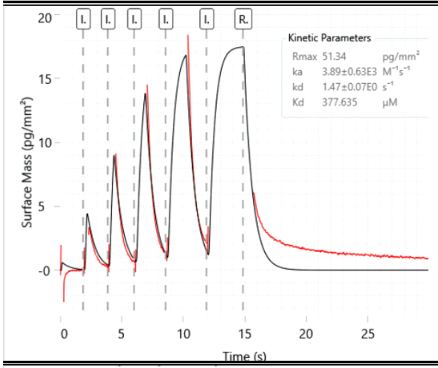
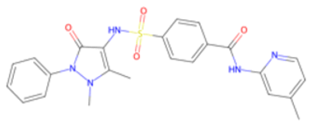
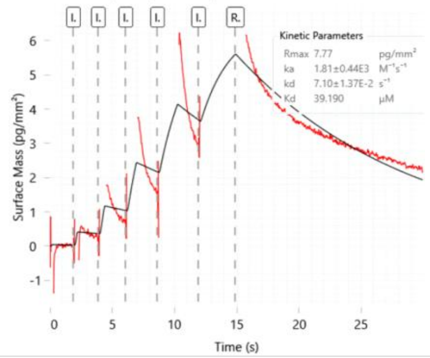
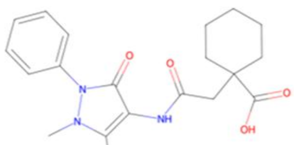
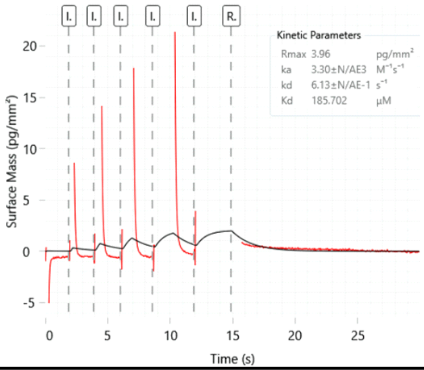
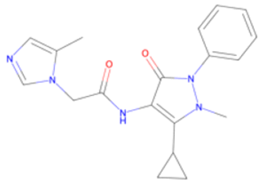
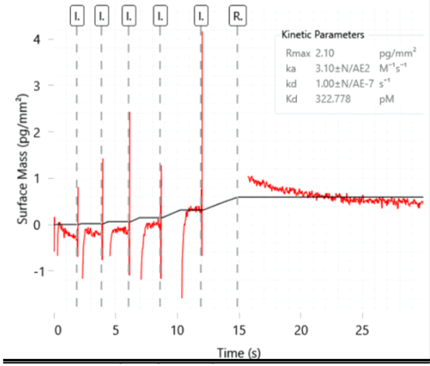
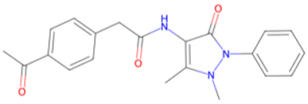
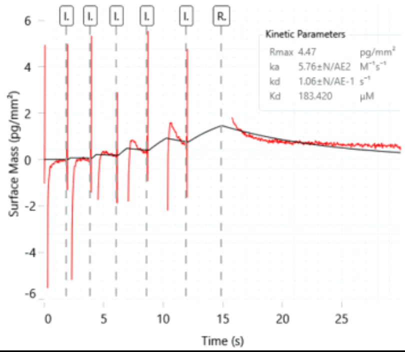
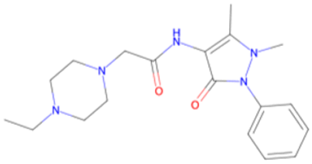
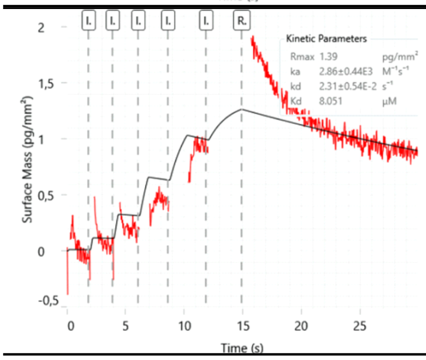
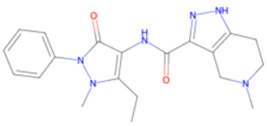
| | | | |
|---|---|---|---|
|  |  SN-2288 – ENAMINE-ID: Z31201539 | KDeep | Kd error: 37%<br>Chi2/Rmax: 42%<br>Rmax/th.Rmax: 40% |
|  |  SN-1242 – ENAMINE-ID: Z126898228 | Kdeep, OnionNet2 | Kd error: 7%<br>Chi2/Rmax: 9%<br>Rmax/th.Rmax: 270% |
|  |  SN-2241 – ENAMINE-ID: Z71389368 | iScore | Kd error: 5%<br>Chi2/Rmax: 4%<br>Rmax/th.Rmax: 500% |
|  |  SN-0709 – ENAMINE-ID: Z169904934 | vScreenMLv2.0 | Kd error: 19%<br>Chi2/Rmax: 18%<br>Rmax/th.Rmax: 100% |

**Figure4.3.2: Binders from single point GCI screening with inaccurate Kd calculation.** Determination of dissociation constant Kd for small molecules fulfilling binder criteria of Rmax above 2 pg/mm$^2$ and Kd below 60% with sensorgrams showing a clear concentration dependent binding response.

GCI measurements of the eight compounds listed in Figure4.3.3 did not display binding characteristics towards PaFabF C164A. Among these compounds, three were from the manual selection.

<table>
<tr><th>Sensorgram.</th><th>Structure:</th><th>Model:</th><th>Kinetic fit markers:</th></tr>
<tr><td>
</td><td><br>SN-1251 – ENAMINE-ID: Z3638453545</td><td>vScreenMLv2.0</td><td>Kd error: 69%<br>Chi2/Rmax: 94%<br>Rmax/th.Rmax: 20%</td></tr>
<tr><td>
</td><td><br>SN-0368 – ENAMINE-ID: Z385088780</td><td>vScreenMLv2.0</td><td>Kd error: 15734271%<br>Chi2/Rmax: 134%<br>Rmax/th.Rmax: 10%</td></tr>
<tr><td>
</td><td><br>SN-0216 – ENAMINE-ID: Z46165720</td><td>Boltz-2</td><td>Kd error: 141%<br>Chi2/Rmax: 74%<br>Rmax/th.Rmax: 20%</td></tr>
<tr><td>
</td><td><br>SN-0846 – ENAMINE-ID: Z4491551051</td><td>OnionNet2</td><td>Kd error: 23%<br>Chi2/Rmax: 18%<br>Rmax/th.Rmax: 10%</td></tr>
</table>

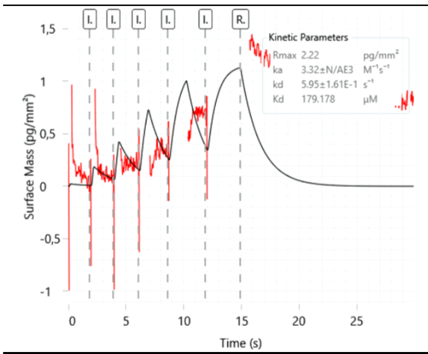
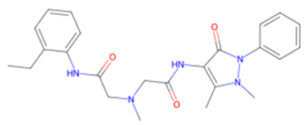
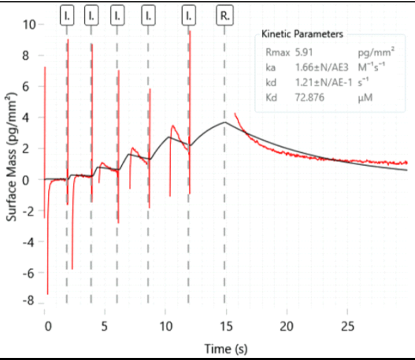
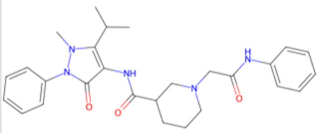
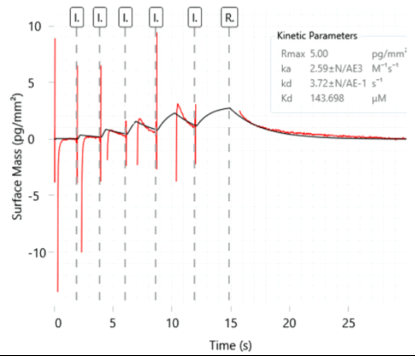
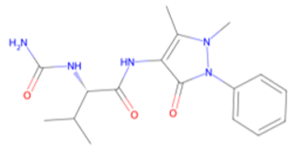
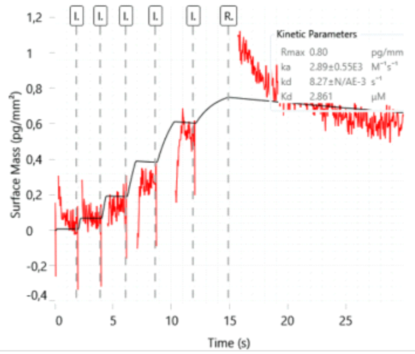
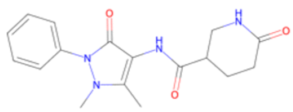
| | | | |
|---|---|---|---|
|  | <br>SN-0073 – ENAMINE-ID: Z219686154 | Manual-selection | Kd error: 27%<br>Chi2/Rmax: 41%<br>Rmax/th.Rmax: 10% |
|  | <br>SN-1442 – ENAMINE-ID: Z5009884376 | OnionNet2 | Kd error: 72%<br>Chi2/Rmax: 96%<br>Rmax/th.Rmax: 20% |
|  | <br>SN-0729 – ENAMINE-ID: Z56875690 | Manual-selection | Kd error: 66%<br>Chi2/Rmax: 61%<br>Rmax/th.Rmax: 70% |
|  | <br>SN-0314 – ENAMINE-ID: Z433859648 | Manual-selection | Kd error: 64%<br>Chi2/Rmax: 42%<br>Rmax/th.Rmax: 10% |

**Figure3.3.1: Measured small molecules where no binding was detected.** Small molecules not fulfilling Rmax>2pg/mm$^2$, Kd<60% and sensorgrams showing a clear concentration dependent binding response.

## 4.3.2 GCI assay summary

Accounting for overlap between selections, Table4.3.2 lists a total of 20 binders identified in the GCI affinity screening. All small molecules from the selections made from AK-Score2, iScore and KDeep top 1% were identified as binders. In contrast, only

2-3 binders were identified from the Boltz-2, OnionNet2, vScreenMLv2.0 and manual selections. All of the small molecules found in more than one selection were identified as binders. The overlapping compounds were SN-1877, SN-0385, SN-1254, SN-1221 and SN-1242. Two compound was not evaluated during the screening by GCI. SN-0163 was not measured to an improper mixing with the protein buffer. SN-0738 was not measured due to a fault during the screening. Sensorgram and kinetic fit parameters of SN-0738 can be found in Appendix (Figure 1)

| AK-Score2 | Boltz-2 | Iscore | KDeep | OnionNet2 | VScreenMLv2.0 | Manual selection |
|---|---|---|---|---|---|---|
| SN-0913 | SN-0163 | SN-2317 | SN-1221 | SN-0366 | SN-0385 | SN-0729 |
| SN-0528 | SN-0216 | SN-1254 | SN-0516 | SN-1242 | SN-0368 | SN-0314 |
| SN-2437 | SN-0953 | SN-2241 | SN-1254 | SN-1442 | SN-0709 | SN-1234 |
| SN-1877 | SN-0738 | SN-1877 | SN-1242 | SN-0746 | SN-1251 | SN-1221 |
| SN-0385 | SN-0156 | SN-2166 | SN-2288 | SN-0846 | SN-0109 | SN-0073 |
| Total identified hits per scoring function selection: | | | | | | |
| 5 | 2 | 5 | 5 | 3 | 3 | 2 |

**Table4.3.2: GCI affinity outcome of the 30 small molecules selected by the six ML scoring functions and manual selection.** Compounds where no binding was detected are highlighted in red, while compounds that could not be evaluated due to experimental failure are highlighted in grey. The total number of identified binders from each selection is shown below.

## 4.4 Validation of binding mode by Xray crystallography

While the GCI affinity assay provided information regarding the strength of the protein-ligand interaction, it did not provide structural insights of how the ligand interacted with the protein binding pocket. Small molecules which displayed high affinity towards PaFabF C164A were subjected to X-ray crystallography in order to explore interactions associated with higher binding affinity. Furthermore, resolved crystal structures would enable evaluation of the initial assumption used during assembly of the ENAMINE-EOS screening set that molecules containing the dimethyl-phenyl-pyrazolone scaffold would adopt the binding mode displayed in Figure1.1.

From the eight compounds co-crystallised with PaFabF C164A, two protein-ligand complexes was resolved by X-ray crystallography. The two complexes included SN-1221 with a resolution of 1.14Å and SN-0709 with a resolution of 1.5Å.

### 4.4.1 SN-1221

X-ray crystallography of SN-1221 (Figure 4.4.1)revealed that the small molecule shared the same dimethyl-phenyl-pyrazolone scaffold binding mode as the 1$^{st}$ generation of the PaFabF C164A EOS series. Additionally the secondary and primary amide of the ligand forms hydrogen bonds with water molecules in the binding pocket.

a) b)

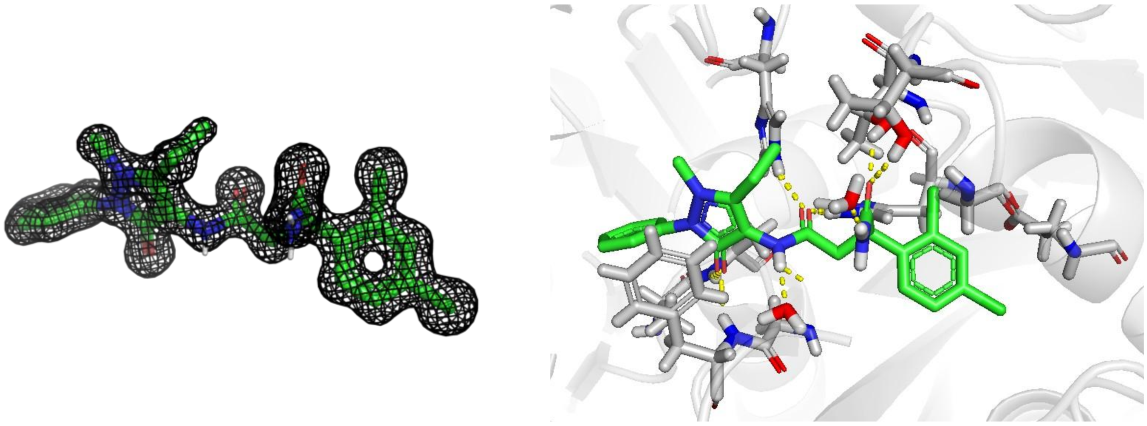


*Figure4.4.1: **SN-1221 Crystal structure.** a) Ligand stick representation of SN-1221 ligand surrounded by the 2Fo-Fc electron-density map contoured at 1.0σ. b) Binding pocket of resolved SN-1221 ligand and PaFabF C164A. Hydrogen bond interaction between protein, SN-1221 and water molecules are displayed as yellow dashed lines. Figure made in Pymol.*

## 4.4.2 SN-0709

X-ray crystallography of the SN-0709-PaFabF C164A complex(Figure4.4.2) show that the small molecule shared the same binding mode as the 1st gen EOS series.

a) b)

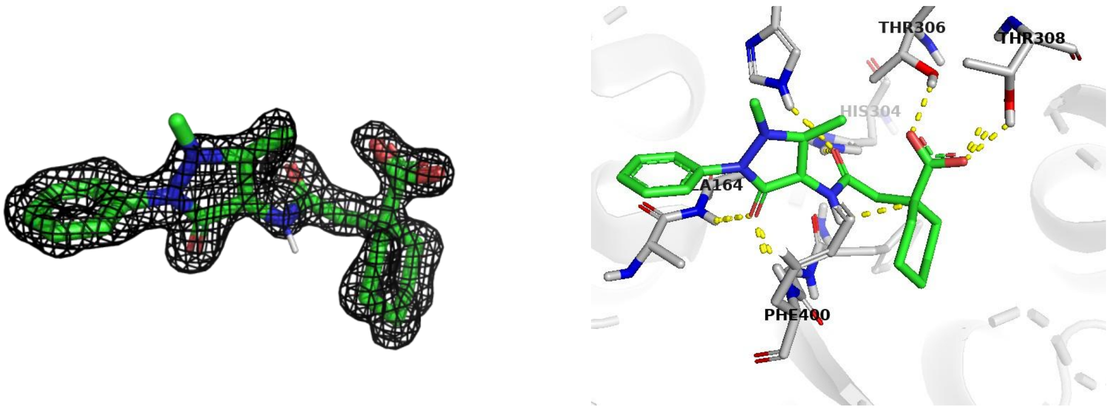



***Figure4.4.2**: **SN-0709 Crystal structure** a) Ligand stick representation of SN-0709 ligand surrounded by the 2Fo-Fc electron-density map contoured at 1.0σ. b) Binding pocket of resolved SN-0709 ligand and PaFabF C164A. Hydrogen bond interaction between protein and SN-0709 are displayed as yellow dashed lines. Figure made in pymol.*

### 4.4.3 Docking power

With the resolved binding pose for two of the screened small molecules in the PaFabF C164A binding pocket, the ability of each scoring function to identify the correct native crystal pose could be evaluated. The docking power of each scoring function was evaluated by measuring the RMSD between the predicted docking pose and the experimentally resolved crystal pose. Listed in Table4.4.3, KDeep predicted the pose with the lowest RMSD of 0.47Å compared to the resolved by X-ray crystal structures of both the SN-1221 and SN-0709 ligands. AK-Score2 predicted the same RMSD value for the SN-0709 crystal pose as KDeep. OnionNet2 predicted the pose with highest RMSD at 1.67Å between SN-0809 and the resolved crystal pose. Boltz-2 predicted the most dissimilar pose between with an RMSD of 1.93Å compared to the crystal SN-1221 pose.

| RMSD calculated between predicted pose and X-ray resolved pose: | | | | | |
|---|---|---|---|---|---|
| | AK-score2 | KDeep | OnionNet2 | vScreenMLv2.0 | Boltz-2 |
| SN-1221 | 1.58Å | 0.47Å | 1.49Å | 1.81Å | 1.93Å |
| SN-0709 | 0.47Å | 0.47Å | 1.67Å | 0.59Å | 1.17Å |

*Table4.4.3: **RMSD between predicted pose and experimentallt resolved X-ray crystal pose.** RMSD values are reported in Ångstrom(Å). Lower RMSD indicate higher similarity between the predicted pose and resolved cryatal pose.*

As seen in Figure4.4.3, the poses selected by KDeep and OnionNet2 kept all interactions made by the crystal SN-1221 ligand. However with the exception of the Boltz-2 generated pose, all the other scoring functions predicted a pose which achieved the same interactions as the 1st gen EOS hit's dimethyl-phenyl-pyrazolone scaffold.

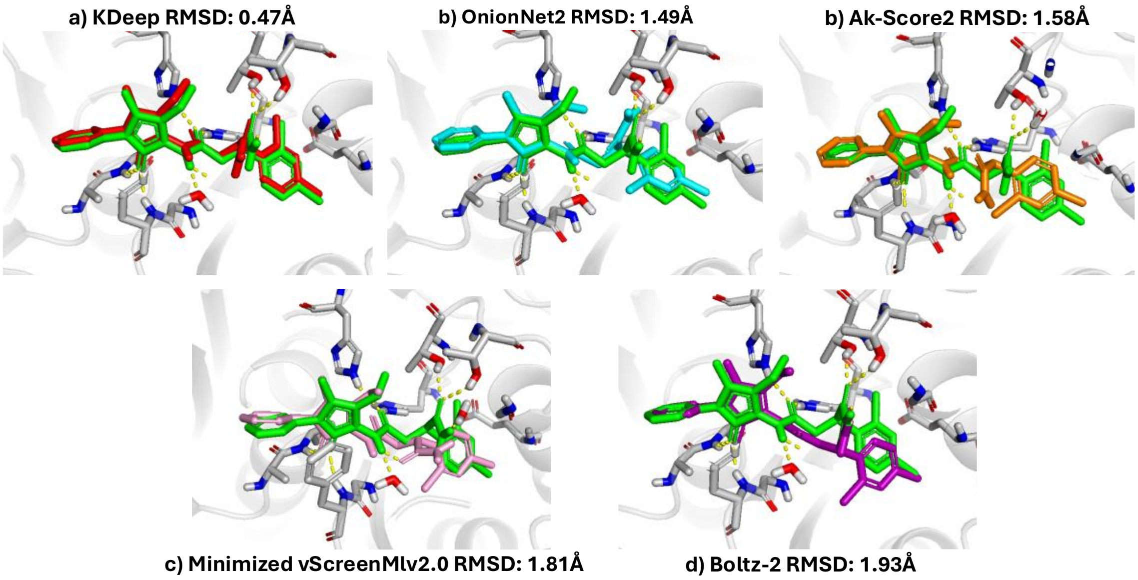



*Figure4.4.3: Comparison of predicted and experimentally resolved binding poses of SN-1221 in the PaFabF C164A binding pocket. a) KDeep selected pose highligted in red, b) OnionNet2 highlighted in cyan, c) Minimized vScreenMLv2.0 highlighten in orange and cofolded pose of d) Boltz-2 highligthed in purple imposed over resolved crystal-structre of SN-1221highlighten in green)and PaFabFC164 complex. The stated RMSD display the*

*difference between the top scored pose and the crystal pose. Figure created with pymol.*

Shown in Figure 4.4.4, for SN-0709 KDeep and AK-Score2 selected poses which shared the same interactions as the crystal pose. OnionNet2 and vScreenMLv2.0 kept the main dimethyl-phenyl-pyrazolone scaffold interactions, but lost the interactions between Thr 306 & 308 and the SN-0709 ligands carboxylic acid. Boltz-2 failed to predict the secondary amide interaction of the dimethyl-phenyl-pyrazolone scaffold.

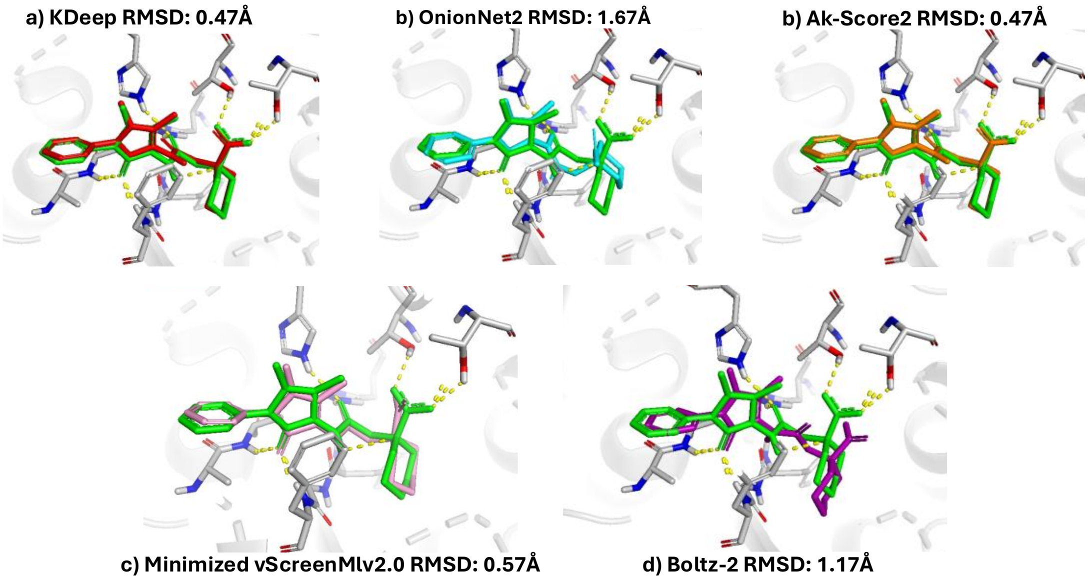



***Figure4.4.4: Comparison of predicted and experimentally resolved binding poses of SN-0709 in the PaFabF C164A binding pocket**. a) KDeep selected pose highligted in red, b) OnionNet2 highlighted in cyan, c) Minimized vScreenMLv2.0 highlighten in orange and cofolded pose of d) Boltz-2 highligthed in purple imposed over resolved crystal-structre of SN-0709 highlighted in green. The stated RMSD display the difference between the top scored pose and the crystal pose. Figure created with pymol.*

For both SN-0709 and SN-1221, vScreenMLv2.0 performed its own minimisation of the generated pose. According to Figure4.4.5, the minimized pose seems to be slightly more similar to the crystal pose for SN-0709.

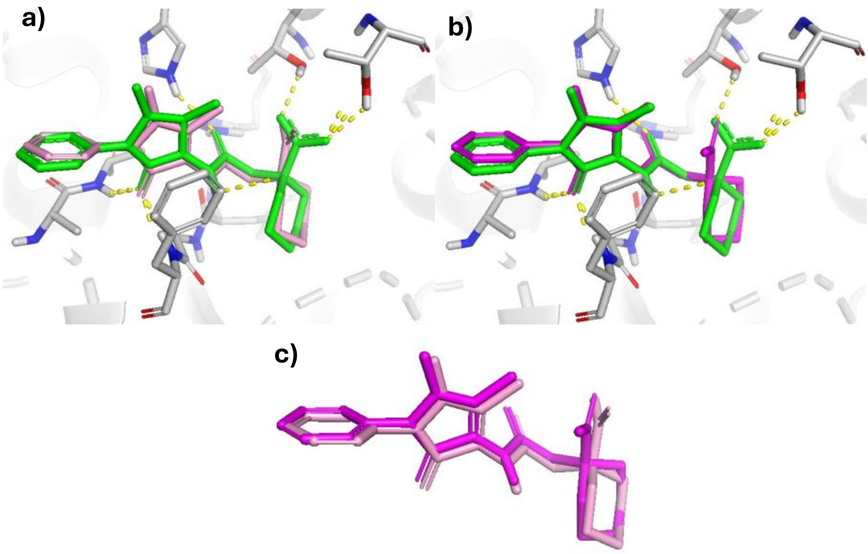



***Figure4.4.5: a) Binding pocket of resolved crystal structure of SN-0709 PaFabF C164A.*** *vScreenML2.0 selected pose after minimization, highlighted in pink imposed over crystal structure of SN-0709 highlighted in green. b) vScreenMlv.20 selected pose before minimization, highlighted in purple imposed over resolved SN-0709 pose highlighted in green. c) Top scored docking pose by vScreenMLv2.0 before minimization imposed over docking pose after vScreenMLv2.0 minimisation.*

# 5. Discussion

To validate the predictions made by the ML scoring functions on the best binder for PaFabF C164A, prospective validation by GCI and X-ray crystallography was performed. To achieve this, a library of compounds containing a scaffold validated to bind the target pocket was assembled. Using the PDB structure of the validated binder 8R0I, docking was performed to generate poses using both GLIDE and FlexX. Of the scoring functions that required a generated pose, these poses was ranked in virtual screening. The scoring functions that did not require a pose performed a virtual screening on the assembled library of compounds. The virtual screening made possible the selection of 5 small molecules from each scoring functions top1% rank. The binding strength of the selected small molecules were evaluated by GCI. Additionally for two compounds, it was possible to resolve the crystal structure of the ligands in complex with PaFabF C164A to validate the binding mode.

## 5. 1 Level of agreement between the six ML-based scoring functions

The Spearman correlation of the pairwise ranking between the six models AK-Score2, Boltz-2, iScore, KDeep, OnionNet2 and vScreen indicate a large variation of how the different models ranked the dataset used for virtual screening. The highest correlation was found between the predicted rank of KDeep and iScore. A Spearman correlation of p=0.58 indicate a moderate agreement between the two scoring functions predicted rank. In other words, on average small molecules achieving a high rank by KDeep's prediction, were mostly ranked high in iScores top rank. The opposite is indicated for the rank produced by iScore and Boltz-2 which received a rank correlation of p=-0.48. Small molecules predicted to be among the strongest binders by Boltz-2, were predicted to be among the weaker binders by iScore. An almost similar negative correlation was achieved between Boltz-2 and KDeep. According to Figure4.1.4d, the most pairwise correlations between the scoring functions ranks was between roughly p=-0.1 and p=0.2. This indicate that there were generally a low correlation between the scoring functions.

An important distinction is that the Spearman correlation was based on the global rank for each scoring function. In virtual screening campaigns, the utility of a scoring function is often determined by its ability to enrich the top of the rank with true binders (Zhou et al., 2024). Thus grading the predictions made on the global rank of the dataset, does not fully coincide primary objectives of scoring functions. On the other hand, global rank correlation provide an insight in how they work. The Spearman corelation heatmap displayed in Figure4.1.4d seem to indicate that the six scoring functions differ in how they predict binding. This indication is further supported by the scatter plots of heavy atom count per compound in relation to the rank position achieved (Figure 4.1.3).

For the ranking performed by the scoring functions KDeep, iScore and OnionNet2, there was a trend between a higher atom count and a higher ranking position. AK-Score2 inherit the same trend, albeit less distinct. The most prominent trend is for the iScore rank. Although less distinct, Boltz-2 have the opposite trend where small molecules with lower atom counts is associated with a higher ranking position compared to small molecules with a higher atom count. A possible cause for the trend of molecules with a high atom count receiving a strong binding prediction could be that for PaFabF C164A, the most potent binders have a high atom count. The relatively higher similarity between the iScore and KDeep rank could further support this claim. However, OnionNet2 which display the same ranking pattern achieve a relatively lower Spearmann correlation compared to the iScore and Kdeep of p=0.06 and p=0.2. AK-Score2 which had a weaker trend for the number of heavy atoms vs rank, had an overall higher pairwise correlation with iScore and KDeep compared to OnionNet2. The AK-Score2, KDeep, iScore and OnionNet2-2 were all trained on the PDBbind dataset. A study investigating the generalization of ML scoring functions found a positive correlation between experimental binding affinity and an increased in buried solvent accessible surface area (SASA) in the PDBbind dataset (Zhu et al., 2022). As the number of heavy atoms is a measurement of molecular size (Kenny, 2019), it could be argued that an increase in heavy atoms will increase the SASA. The cause for the trends shown in (Figure4.1.3 c, e, f), could be that the models trained on PDBbind has learned a pattern of favouring larger molecules.

In this study, only selected molecules of the top 1% was validated prospectively. Figure4.1 show that there is some degree of overlap between the scoring functions in their top 1% rank. The overlap of the top5% and 25% ranks was calculated to check the level of agreement in other relatively higher sections of the ranks. The Figure 4.1.4a and b show a generally low level of overlap compared to what would be expected by random. It seems like the pattern of similarity between the ranks stayed consistent. Disagreement between scoring functions is not unique to ML-based approaches. Previous benchmark studies of traditional scoring functions has reported variability in how benchmark sets are ranked (Gregory L. Warren et al., 2006; Perola et al., 2004; Zhou et al., 2007). However, the level of agreement is not universal between the six ML-based scoring functions. Similar to the global rank correlation shown in Figure4.1.4 d, iScore and KDeep both achieve the highest similarity in the two similarity heatmaps of Figure 4.1.4 a and b.

Furthermore, in the heatmap of Figure 4.1.4c, KDeep and iScore achieve a relatively similar prediction compared to the other scoring functions for their top 1% ranked compounds. Boltz-2 on the other hand which had the opposite trend of predicting molecules with lower atom count as the most potent, achieved a low median percentile ranking of between 94.2% and 70% in the ranks predicted by OnionNet2, Kdeep and iScore.

### 5.1.1 Retrospective validation of the 1$^{st}$ gen EOS hits

The rescoring of the three 1$^{st}$ generation EOS hits show that none of the scoring functions were able to properly identify these small molecules as strong binders. As these hits achieved Kd in the low micromolar range, they should have achieved a high rank. Kdeep was the only scoring function to rank one of the EOS hits in the top 10%. However, KDeep also displayed the most dispersion by ranking the other two hits in the bottom 20$^{th}$ percentile. The 8R1V ligand was also the small molecule with the highest atom count of 27 among the three hits. The other two had an heavy atom count of 24 and 21. As the Hit 1 had a slightly worse Kd at 23μM compared to Hit 2 with a Kd of 19μM, it should be ranked slightly higher with far less dispersion.

The approximate mean ranking of the three EOS hits shown in Figure4.2b also vary a substantial amount among the six scoring functions. This show that the ensemble ranking of all six scoring functions was not in agreement for the 1$^{st}$ gen EOS hits. A noteworthy observation is that the two models which did not require a pose, had almost the smallest difference in the dispersion of ranking between the three hits, only being surpassed by vScreenMLv2.0. Boltz 2 which did its own co-folding of the ligand inside the binding pocket also achieved the best mean rank of the three compound. Additionally as vScreenMLv2.0 performs a minimization of the protein-ligand complex which slightly alters the ligand pose, it does not rely as heavily on the generated pose as the other scoring functions. A counter for this argument was that the redocking of these hits only had an RMSD ranging from 0.34Å to 0.88Å compared to the native pose..

### 5.1.2Physiochemical variation in the selection of the top 1% rank

The 5 selected compound from each scoring function shown in Figure 4.3.1, 4.3.2 and 4.3.3, provide some insights into which chemical features each scoring function predicted to be important for binding. Of the vScreenMLv2.0 selection, the compounds seem to be generally polar. The selection also only contained the dimethyl-phenyl-pyrazolone amide scaffold. Kdeep differed by selecting larger compounds. This also coincided with KDeeps tendency to favour larger molecules within its top rank

(Figure4.1.3b). Two of the small molecules in the KDeep selection contained hydrophobic cages. The selection also contained one dimethyl-phenyl-pyrazolone sulphonamide scaffold. In the AK-Score2 selection, there was a total of two small molecules with a sulphonamides. Both of the sulphonamide small molecule contained nitro groups. In general, the AK-Score2 selection consisted of relatively medium sized polar compounds. Boltz-2 on the other hand selected no small molecules with the dimethyl-phenyl-pyrazolone sulphonamide scaffold. The Boltz-2 selection generally contained polar molecules with a smaller size compared to the selections from the other scoring functions. This correlate with the scatter plot of heavy atom count versus rank for Boltz-2 Figure4.1.3d. Of the OnionNet2 selection, there were no dimethyl-phenyl-pyrazolone sulphonamide scaffolds. The molecules were of relatively larger size compared to Boltz-2. Both the SN-0746 and SN-1242 compounds in the OnioNet selection contained hydrophobic cages. The SN-1221 was also found in the KDeep selection.

The iScore selection was distinct in the sense that four out of 5 compounds contained the sulphonamide dimethyl-phenyl-pyrazolone scaffold. As only roughly 20% of the dataset consisted of sulphonamides, iScore seems to strongly favour this chemichal group. iScore was the only model which did not use poses for scoring. Furthermore, the three 1$^{st}$ gen EOS hits which served as the template for pose generation did not have the dimethyl-phenyl-pyrazolone sulphonamide scaffold. Thus, docking small molecules containing sulphonamides with a docking grid based on a amide scaffold, could provide the scoring functions with poses that were more dissimilar to the true native pose. A counterargument for this would be the lack of dimethyl-phenyl-pyrazolone sulphonamide scaffolds in the Boltz-2 selection as Boltz-2 did not rely on a amide scaffold template for its generation of a pose. Overall, the iScore selection contained molecules of a relatively larger size. The two compounds SN-1254 and SN-1266 contained indications of hydrophobic scaffolds.

## 5.2 Scoring functions ability to identify binders for PaFabF C164A

From the screening of the selected small molecules affinity towards PaFabF C164A by GCI, 20 ligands was identified as binders (Table 4.3.2). Among the identified binders, the two compounds SN-1221 and SN-0385 achieved a confident Kd prediction in the low micromolar range. SN-1221 with the lowest Kd was not found in the selection for the other scoring functions. However, SN-1221 was selected during the visual inspection of the GLIDE generated poses. SN-0385 was selected by vScreenMLv2.0 and AK-Score2. All of the overlapping compounds in the selections showed binding characteristics.

Among the molecules with a confident Kd prediction, the small molecules with the highest Kd SN-1221 and SN-0385 were similar in the sense of not being too large and a sharing a relatively similar polar composition. However besides sharing the same dimethyl-phenyl-pyrazolone amide scaffold, they did not share other similar chemical groups. For the binders there was less correlation between the molecular scaffolds. Both binders with hydrophobic cages and a larger size like SN-1254, and smaller molecules like SN-0528.

Of the 20 identified binders, 6 contained sulphonamides. This indicates that sulphonamide containing analogues were capable of copying the interaction between the secondary amide and the two histidine's of the malonyl binding pocket (Figure 1.1). However, this was not verified in this study as attempts of resolving the binding mode of two sulphonamide containing compounds SN-1877 and SN-2288 by X-ray crystallography failed.

As shown in Table 4.3.2, all compounds from the selections of the three scoring functions KDeep, AK-Score2 and iScore were identified as binders. A previous prospective validation of AK-Score2 also identified it as capable of identifying binders (Hong et al., 2024). The validation performed in this study further validate AK-Score2 capabilities of identifying binders. The precursor vScreenML of vScrenMLv2.0, has also been used to identify novel binders with success (Adeshina et al., 2020). The ability of vScreenMLv2.0 to identify one ligand with an accurate low micromolar affinity, and one binder with resolved binding mode prospectively validates vScreenMLv2.0 as a scoring function able to discover novel binders.

Neither Keep or iScore has previously been prospectively validated. However a noteworthy observation is that these models overall displayed the highest similarity. This indicates that they  identified the same underlying pattern for a strong binding between the ENAMINE-EOS set and the malonyl binding pocket of PaFabF C164A. IScore and KDeep employed different ML architectures, but they were trained on similar data. The higher level of similar prediction compared to the other ML-based scoring functions could indicate that for ML-based scoring function, the training data matter more than architecture and data input. On the contrary, OnionNet2-2 which also seemed to predict relatively similar to iScore and KDeep while also being trained on the same dataset was not able to identify as many binders. Despite no measurement for two of the binders in the Boltz-2 selection, Boltz-2 was still able to find binders. This was interesting as Boltz-2 top1% ranked molecules had a median percentile rank close to the bottom ranks for iScore and KDeep. It could indicate that the ENAMINE-EOS library contained many binders.

From the binding data retrieved from the single point GCI screening of the 28 compounds (Table4.3.2) against PaFaBF C164A, KDeep and AK-Score2  was able to retrieve the strongest predicted binders. The ability to both identify binders in the low micromolar range while not contain any non-binders is unique for KDeep and AK-Score. For this screening, AK-Score2 and KDeep was the best performing ML-based scoring functions.

## 5.3 Validation of predicted binding mode

The resolved structures of SN-1221 and SN-0709 further validates these small molecules as binders. Both ligands shared the same main binding mode as the 1st generation EOS series which further validates the assumptions that the use of the dimethyl-phenyl-pyrazolone amide scaffold for finding other hit compounds. It also indicate that the scoring functions KDeep and vScreenMLv2.0 were able to interpret this binding mode as a contributor to a strong binding affinity. Interestingly, the crystal binding pose of SN-1221 made two additional new hydrogen bonds with two waters in the binding pocket compared to the generated pose. The hydrogen bond interaction with a water molecule and the secondary amide hydrogen of the 1st gen EOS dimethyl-phenyl-pyrazolone scaffold, is also present in the crystal binding mode of Hit1 shown in Figure1.1. However in the preparation of the 8R0I for docking of the ENAMINE-EOS set by GLIDE, this water molecules was not included. Both KDeep and the visual inspection predicted this compound to have a high affinity even though this water molecule was not accounted for in the generated pose. The hydrogen bond interaction with the primary amide is a new interaction not present in the three 1st gen EOS hits. Both the SN-1221 and SN-0709 made interactions with the Threonin 306 and 308 which was set as rotable for docking. This interaction is not observed for any of the 1st gen EOS hits, but has been observed for the 7OC1 crystal structure of the potent binder PTM in complex with PaFabFC164Q.

### 5.3.1Docking power

A metric for determining the accuracy for a scoring function is its ability to select a pose close to the native crystal pose (Spyrakis et al., 2007). A common benchmark criterion for docking power is a predicted pose with an RMSD of ≤2 Å relative to the native crystal pose(Schaller et al., 2024). The calculated RMSD between the top selected pose for SN-1221 for the five scoring functions (Table3.3,Figure4.1), show that all scoring functions was within the 2Å mark. Figure4.1 further validate that most the selected poses did not deviate much from the crystal pose. The exception was Boltz-2 generating a pose which did not achieve hydrogen bond interaction with the two Histidine's 341 and 304. For the prediction of the binding mode for ligand SN-0709, the trend was similar. All of the scoring functions which predicted based on a pose, selected poses within the 2Å RMSD benchmark. Boltz-2 did not predict a histidine binding for the secondary amide. The pattern of Boltz-2 failing to predict the validated binding mode of the dimethyl-phenyl-pyrazolone scaffold coincide with its inability to predict this interaction for its top 1% predicted compounds. Interestingly, KDeep only predicted this interaction for roughly 40% of its top 1% predicted small molecules. This observation does not support that KDeep fully identified this binding mode as important for binding for binding affinity. vScreenMLv2.0 on the other hand was able to identify this binding mode for all of its top 1% ranked small molecules. For this study, vScreenMLv2.0 seems to have the highest docking power.

The comparative assessment of scoring functions (CASF-2013) found that generally the docking and screening power of traditional scoring functions tend to be better than their scoring and ranking capabilities. Furthermore, the study recognised that the best performing scoring functions achieved success rates above 80% for selecting the pose within the 2Å cutoff (Li et al., 2014). As vScreenMLv2.0 show the highest potential for identifying correct poses, it would be interesting to further asses vScreenMLv2.0s docking capabilities outside its top1% rank. What made vScreenMLv2.0 distinct from the other scoring functions was its built in minimization of the generated pose. Although not much, it seems like the minimised pose was moved closer to Thr308 for SN-0709 (Figure4.4.5). Perhaps it was the minimisation performed by vScreenMLv2.0 on all the poses which abled vScreenMLv2.0 to identify the validated binding mode as important for binding.

Based on the low agreement between scoring functions and their inability to properly rank the 1st gen EOS series, their ranking capabilities seems to be limited. However, due to their ability to identify binders and retain known binding modes for SN-1221 and SN-0709, there is optimism for their abilities in screening and docking power. This observation is consistent with the findings of the CASF-2013 regarding traditional scoring functions suggesting that ML-based scoring functions struggle with the same challenges. As scoring power was not evaluated in this study, no conclusions can be drawn.

## 5. 4 Limitations

The six scoring functions was only tested towards one single target using a curated dataset with compounds containing a scaffold known to bind this target. This limits the evaluation of the performance of the scoring functions towards only this target where there is already substantial information of the mechanisms for binding. Consequently, the observed performance of the scoring functions may not be generalizable to other targets. The next challenge for these ML based scoring functions would be to utilise them to predict novel binders for different targets with larger and less curated chemical libraries. On the other hand, the six ML based scoring functions could also be implemented to further explore the malonyl binding pocket as they have proven to find binders which later revealed both novel and previously observed interactions in the binding pocket.

The evaluation of docking power was only done on the scoring functions ability to predict a pose close to the SN-0709 and SN-1221 ligands. Rescoring poses of the 1st gen EOS series would provide a better assessment of the docking power for each scoring function.

The relatively strong affinity of SN-1221 and SN-0385 towards PaFabF C164A identified them as potent binders. Examining their ranking positions in the other ML-based scoring functions could provide new insights on the ability of the scoring functions to identify potent binders.

In addition to attempt X-ray crystallography on the ligands with a predicted Kd below 100μM, it was decided to also include the compound SN-0163 as it was not subjected to GCI screening. At the time of the decision, it was believed that the measurement of SN-0738 displayed the characteristics of an insoluble ligand. Later by closer inspection of the kinetic parameters and sensorgram, it was concluded that the measurement itself failed. This can be seen in the appendix (Figure1). By the reasoning for why SN-0163 was subjected to X-ray crystallisation, SN-0738 should therefore also have been subjected to X-ray crystallography.

Increasing the number of selected compound from each rank for prospective validation, could increase the accuracy of determining the ability for each scoring function to identify binders.

The complexity and difficulty of using the ML based scoring functions, is a limitation. Most of the scoring functions required specialised environments and extensive data preprocessing. his restricts the use of these models to users with a certain level of computational expertise, thus limiting broader application.

A key limitation was only screening these compounds against the PaFabFCA164A once. The single screening by GCI resulted in a general uncertainty towards most of the compounds in the selection. It mostly served as an indication towards whether the small molecules displayed binding properties. The addition of crystals structures for two of the ligands validated them as true binders toward the target. A new attempt on X-ray crystallisation for all of the selected compounds could potentially reveal new binders and binding modes. The six ML scoring functions varied substantially in their global ranking of the library. It could be interesting to find the true number of binders in the ENAMINE-EOS library by screening all 1368 compounds. Measuring the affinity for all the compounds would provide a more accurate assessment for the ability of each scoring function to find potent binders. Perhaps the ENAMINE-EOS contain more potent binders which the scoring functions failed to identify.

## 5.5 Conclusion

To conclude, the virtual screening enabled the selection of 30 compounds from a library consisting of 1368 compounds. The selection consisted of 5 compounds from the top 1% strongest predicted binders for each of the six scoring functions iScore, OnionNet2, AK-Score2, vScreenML and KDeep. In addition, 5 compounds were selected by visual inspection of the generated poses. Chemical properties such as polarity, functional groups and size varied between the selected compounds indicating that the different scoring functions favoured different chemical features in the prediction of a strong binder. A possible testament to their difference in training and ML architecture. After an *in vivo* screening by Grafting Coupled Interferometry (GCI), a total of 20 compound displayed binding properties towards PaFabF C164A. Of the 20 identified binders, two achieved a fairly confident low micromolar Kd prediction. X-ray crystallography of the binder with the lowest micromolar prediction further validated the binder while also revealing novel interactions in the PaFabF C164A malonyl binding pocket. In addition to the predicted strongest binder, another binder was confirmed by X-ray crystallography. All of the ML based scoring functions which relied on a pose were able to predict a pose within a 2Å RMSD difference between the crystal poses of the two resolved ligands. However, despite achieving a RMSD below 2Å, Boltz-2 failed to generate a pose that followed all key interactions validated for the dimethyl-phenyl-pyrazolone scaffold. Due to their ability to exclusively select binders, identify two compounds in a low micromolar affinity range and predict poses with the experimentally validated binding mode, AK-Score2 and KDeep exhibited the strongest overall performance among the evaluated scoring functions. Furthermore all of the compounds which shared overlap in the selection by other scoring functions were identified as possible binders.

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